ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

INDEX - MACHINE HISTORY

MACHINE NUMBER : MU-719

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	MACHINE DISPATCH CLEARANCE	F-QA-FA -16	✓
2	PACKING LIST / DISPATCH DETAILS	F-TSG-18 & F-PAC-01	✓
3	MINUTES OF MEETING	F-TSG-17	—
4	WORK ORDER	F-TSG-15	✓
5	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
6	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
7	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
8	CHECK LIST - MECHANICAL	F-QA-FA-07	✓
9	MACHINE HISTORY - MECHANICAL	F-QA-FA-10	✓
10	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
11	MACHINE HISTORY - ELECTRICAL	F-QA-FA-14	✓
12	FINAL ASSEMBLY REPORT - MECHANICAL	F-QA-FA-05/15	✓
13	GEOMETRICAL TEST REPORT	DS-DSN-PROD-TC 01~ 06	✓
14	PRACTICAL TEST REPORT	F-TSG -34	✓
15	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
16	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
17	HYDRAULIC ASSEMBLY REPORT (OPTIONAL)	F-QA-IPR-HYD-01	—
18	BALL BAR REPORT	F-QA-FA-02	✓
19	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
20	GEAR BOX / ATC ASSEMBLY	F-QA-IPR-GBA- 01 / SUPPLIER REPORT	✓
21	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	✓
22	BASE BUILDING	F-QA-IPR-BBA- 01/ 02	✓

Date : 28/9/11 CHECKED BY APPROVED BY

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MACHINE DISPATCH CLEARANCE

M/c No.: MH-719	CUSTOMER : M/S MATHARU ENGINEERING WORKS
MODEL : MCV-400	DATE : 30/9/11

DEVIATION DETAILS

QA

ASSEMBLY IN CHARGE

QUALITY IN CHARGE

AUTHORIZED SIGNATURE

CORRECTIVE ACTION (BY CSG)

RESPONSIBLE PERSON :

SIGNATURE OF CSG HOD

Note : Customer Support Group (CSG) will confirm the closure of deviation and Return to QA.

600

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MACHINE DISPATCH CHECK POINTS

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CUSTOMER'S NAME: Maharu Engineering Works Faridabad

M/C No.: M11-719 MCV-400 DATE: 03-10-2011 Packed

S.No	DESCRIPTION	YES/ NO/NA	S.No.	DESCRIPTION	YES/ NO/NA
1	Packing list available	Yes	21	Chip Trolley cleaned	NA
2	Hydraulic / Pneumatic Counter Balance Cylinder removed	NA	22	Standard Spares & m/c manual available	Yes
3	Z- Motor Cover Removed & Packed	NO	23	Coolant tank assly. packed	Yes
4	Side door handles removed & Packed	NO	24	Index Table Cover packed properly	NA
5	Base guard removed & Packed	Yes	25	Oil skimmer bin cleaned	NA
6	Coolant Gun tied properly	Yes	26	Voltage Stabilizer packed	NA
7	Air gun tied properly	NA	27	Machine Platform cleaned	NA
8	Leveling Screws, Nuts removed & Packed	Yes	28	Wanner Pump & cover cleaned & put back	NA
9	Leveling blocks 100/180/220/300 Ht. Qty: - 4/6/7/8/11/20	Yes	29	Customer components packed properly	NA
10	Automatic Tool Changer	Yes	30	Index Table packed properly	NA
11	APC Assembly packed Properly	NA	31	Chiller unit cleaned	NA
12	Pallet removed & Packed	NA	32	Rotary Table packed properly	NA
13	APC leveling screws - 1set	NA	33	Toolings packed properly	NA
14	Power pack stand	NA	34	Tail Stock packed properly	NA
15	Lub Tank oil drained - retain 2 ltrs.	Yes	35	Practical Test piece packed	NA
16	Machine oil Bin cleaned	Yes	36	Fixture/s packed properly	NA
17	Coolant tank mesh cleaned & put back	Yes	37	Air removed from cover with Vaccum cleaner	NA
18	Chip Bin cleaned & put back	Yes	38	Desiccant Bags Hung	NA
19	Chip Bin mesh cleaned & put back	Yes	39	Fumigation Certificate available	NA
20	TPM points stickers fixed properly	NA	40		

REMARKS / SHORTAGES IF ANY:

ASSEMBLY	STORES	FIXTURE	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY

Jayanta.

F-PAC-01-RO

ACE MANUFACTURING SYSTEMS LTD.

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DISPATCH CHECK POINTS

CUSTOMER'S NAME : Matharu Engineering Works Faridabad

MACHINE NUMBER : M11-719 MCV-400 **DATE :** 03-10-2011

S.No	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	Yes	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	Yes	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE Inside the cabinet	Yes	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON M/c	Yes	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	NO	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	Yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	CHECK ITEMS AS PER PACKING LIST	Yes	
22	WOODEN PACKING PROVIDED BELOW THE APC STATION.	NO.	
23	WOODEN Support given below the Grundfoss pump Mounting bracket	——	this proposal Done by 23/4/2011
24.	grundfoss pump Removed and packed seperately	Yes/NA	
25	MACHINE LIFTING PROCEDURE SKTCH PASTED ON THE GUARD	Yes	

ATTENDED BY | QUALITY ASSURANCE | DISPATCH - INCHARGE

F-PAC-02-R0

CONTROLLED COPY

Sign: M Date: 1/8

ACE MANUFACTURING SYSTEMS LTD

PACKING CASE IDENTIFICATION LIST

AMS

MACHINE NUMBER : M11-719 DATE : 03-10-2011

SL.NO	CASE - 1 CONTENTS	QUANTITY
1	Machine assembly	01NO
2	Std spares and M/c Manuals	01Box
3	Leveling screws & Lock Nut	4+4Nos
4	Leveling Block 150MM	08Nos
5	Spin-Rail	01L2+08
6	Coolant gun	01NO
7		
8		
9		
10		
11		
12		

CASE - 2 CONTENTS

1	Coolant tank assembly	01NO
2	Coolant Pump	04Nos
3	Chip bin	01NO
4	chip bin Mesh	02Nos
5	chuck ear plate	01NO
6	Oil bin	01NO
7	base gaurd	01NO
8	Mesh	08Nos
9		
10		
11		
12		

CASE-3 CONTENTS

1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		

CHECKED BY : Dhanus

ACE MANUFACTURING SYSTEMS LTD. 514 M/C.Sl. No. M11-719

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PACKING LIST ANNEXTURE-II

M/s, Mathary Engineering Works, Faridabad

Sl.No.	DESCRIPTION	QTY.
1	O'Ring Dia 19.1X1.6	2
2	O' Ring Dia 8.1X1.6	6
3	O' RingDia 11.1X1.6	4
4	O'Ring Dia 57.6X2.4	6
5	O'Ring Dia 24.5x3.0	6
6	O'Ring Dia 51.6x2.4	2
7	Cord Of O'Ring Dia 3 X 2 Mtrs	1
8	Spanner 10 x 11	1
9	Spanner 12 x 13	1
10	Spanner 18 x 19	1
11	Spanner 30 x 32	1
12	Spanner 36 x 41	1
13	Allen key set Metric.	1
14	Extension Allen Key **8mm**	1
15	Split Collet With Stud Push	2
16	Fuse Set (Fanuc Make)	1
17	Glass Fuse Set (Kept in Electrical Cabinet)	1
18	IML-Process Oil - 12 (Kept in Machine) **Servo Spin-12**	1
19	Pull Stud BT-40 (**DIN**,MAS 403)	20
20	Fanuc Series Oi Mate Model D / Oi-Model D **USER`S MANUAL**	1
21	Fanuc Series Oi-Model D / Oi Mate Model D **MAINTENANCE MANUAL**	1
22	Fanuc AC SERVO & SPINDLE MOTOR **Bis series /~~alpha series~~** MAINTENANCE MANUAL	1
23	Operating Instructions And Maintainance Manual	1
24	Spare Parts Manual	1
25	Air Cooler Manual	1
26		
27		
28		
29		
30		
31		
32		
33		
34		

Ace Micromatic Group Company

Sign.....

DATE 30-09-11

F-STR-27-R0

ACE MANUFACTURING SYSTEMS LTD.,

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PACKING LIST ANNEXTURE-III

M/C.Sl. No.MS-119

Customer:M/s, Madhava Engineering Works, [illegible]

Sl.No.	DESCRIPTION	QTY.
1	ELECTRICAL MANUAL	1
2	MACHINE BACKUP - CD	1
3	FANUC SYSTEM GUARRANTY CARD	1
4		
5		
6		
7.		

Sign.....

DATE.....02.03.2011

ACE MANUFACTURING SYSTEMS LIMITED		AMS In Pursuit of Excellence
PACKING LIST		
PACKING LIST NO.	DNo#005385	Date: 29/09/2011
SALES ORDER NO.	MO10001571	Date: 25/06/2011
CUSTOMER	MATHARU ENGINEERING WORKS., FARIDABAD	
P.O. NO.	NIL Date : 13/06/2011	

Sl. No.	Item Id	Model	Qty
1	M11-719	MCV-400	1.000

Sl. No.	Description	Short Supply ?
1	CNC Vertical Machining Center Model "MCV-400"	
	Spindle nose taper 7/24 No.40	
	Spindle power (Fanuc) 15 Min./Cont. Rating - 7.5 / 5.5 KW	
	AC Servo axes drive (Fanuc)	
	20 Station tool magazine	
	Twin Arm (Random) type High Speed Auto tool changer	
	Tool shank BT-40	
	LM Guideways for all the three axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ball screw stretching for X & Y Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Standard CNC features of Fanuc 0IMATE-MD with Colour LCD	
	Spindle speed of 60 - 6000 rpm (Indirect drive)	
	Rapid traverse of 30 / 30 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Flood coolant system around the spindle	
3	Chip flushing system on the base	
4	Coolant gun	
5	Pull Studs (20 nos.)	
6	SERVO SPIN - 12 lubricating oil (1 liter)	
7	1 set of glass fuses(inside the cabinet)	
8	Electrical documentation manual & back up CD	
9	Geometrical test report	
10	NAS test report	Inside the cabinet
11	Laser Calibration test report	
12	Heavy cut & Rigid tapping test report	

ACE MANUFACTURING SYSTEMS LIMITED.

DISPATCH ADVICE

AMS
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Sl. No.	Description Of Dispatch details	
1	Customer Name	MATHARU ENGINEERING WORKS., [Party Id - MATEN006]
2	Telephone No.	
3	Fax No.	
4	E-Mail	
5	Customers ECC No.	ACQPS8886AXM002
6	Customers Sales Tax No. & Date	CST: TIN: 06921308561
7	Sales Order no.	MO10001571
8	Machine Model	MCV-400
9	Machine No.	M11-719 [MACHINE UNDER EPCG SCHEME]
10	Invoice Address	MATHARU ENGINEERING WORKS., PLOT NO.30, III SUB DIVISION, WEAR WELL CYCLE COMPLEX, INDUSTRIAL AREA, N.I.T, FARIDABAD
11	Place of Delivery	Default Address
12	Purchase Order No. & Date	NIL Date:13/06/2011
13	P.O Value (Excluding Taxes & Duties)	Rs.20,40,000.000
14	Transportation	Door delivery through any reliable carriers and freight to-pay basis
15	Insurance	To be arranged by AMS and the charges to be included in the Invoice. Dispatch details to be faxed to the customer.
16	Scheduled Dispatch	04/10/2011
17	Payment	Refer payment collection detail sheet
18	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch Shortage list to be made while preparing machine invoice All Pending Points to be attended before machine dispatch

NOTE: ENCLOSED ROAD PERMIT

Copy: Production/Electrical/Design/Purchase/Accounts/Dispatch/TSG

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST

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PACKING LIST NO.	DNo#005385	Date: 29/09/2011
SALES ORDER NO.	MO10001571	Date: 25/06/2011
CUSTOMER	MATHARU ENGINEERING WORKS., FARIDABAD	
P.O. NO.	NIL Date : 13/06/2011	

Sl. No.	Item Id	Model	Qty
1	M11-719	MCV-400	1.000

Sl. No.	Description	Short Supply ?
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	Spindle nose taper 7/24 No.40	
	Spindle power (Fanuc) 15 Min./Cont. Rating - 7.5 / 5.5 KW	
	AC Servo axes drive (Fanuc)	
	20 Station tool magazine	
	Twin Arm (Random) type High Speed Auto tool changer	
	Tool shank BT-40	
	LM Guideways for all the three axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ball screw stretching for X & Y Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Standard CNC features of Fanuc 0IMATE-MD with Colour LCD	
	Spindle speed of 60 - 6000 rpm (Indirect drive)	
	Rapid traverse of 30 / 30 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Flood coolant system around the spindle	
3	Chip flushing system on the base	
4	Coolant gun	
5	Pull Studs (20 nos.)	
6	SERVO SPIN - 12 lubricating oil (1 liter)	
7	1 set of glass fuses(inside the cabinet)	
8	Electrical documentation manual & back up CD	
9	Geometrical test report	
10	NAS test report	
11	Laser Calibration test report	
12	Heavy cut & Rigid tapping test report	

ACE MANUFACTURING SYSTEMS LIMITED

WORK ORDER

AMS
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Customer : **M/s MATHARU ENGINEERING WORKS.,** PLOT NO.30, III SUB DIVISION, WEAR WELL CYCLE COMPLEX, INDUSTRIAL AREA, N.I.T, FARIDABAD HARYANA	**SO. No:** MO10001571	**WO. Date :** 07/07/2011
	PO. No.: NIL	**PO. Date :** 13/06/2011
	S.O.TYPE: Standard	

Sl. No	Item Id	Item Description	Unit	Qty
1	VMC	CNC VERTICAL MACHINING CENTER - MCV-400	Nos	1

MACHINE COLOUR :

Machine Colour : ACE Blue & Light Grey combination.

WARRANTY :

a) One year (i.e. 12 months) from the date of commissioning of the machine or 15 months from the date of supply of the machine whichever is earlier against faulty material and workmanship. Guarantee is not applicable for part, which has normal wear & tear (like spings, balls etc.) and certain other items like Bulbs, Lamps, Indicator Lamps, Revolving Lamps, Seals and Glass Fuses.

b) For Fanuc CNC System, motors and drives, it is Two Years from the date of despatch. This warranty is applicable only for the first installation of the machine.

Scope of Supply, Technical Specs & CNC Features As Per **Annexure I**

Machine Inspection	Geometrical accuracies and machine acceptable as per test chart and NAS test
Customer Inspection Type	Direct
Customer Inspection Date	09/11/2011
Component Trials Required?	Not Applicable

FOR ACE MANUFACTURING SYSTEMS LTD.,

(P.RAMADAS)
MANAGING DIRECTOR

(L.S.UMESH)
CHIEF EXECUTIVE OFFICER

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

AMS — In Pursuit of Excellence

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
1	TABLE		
1.1	TABLE SIZE	mm	900 x 450
1.2	T-SLOT DIM.	N x W x P	4 x 18 x 100
1.3	MAX. LOAD ON TABLE	Kgf	300
1.4	DIST. FROM FLOOR TO TABLE TOP	mm	925
2	MACHINE CAPACITY		
2.1	X - AXIS TRAVEL	mm	600
2.2	Y - AXIS TRAVEL	mm	450
2.3	Z - AXIS TRAVEL	mm	500
2.4	DIST. FROM SPINDLE FACE TO TABLE TOP	mm	175 - 675
2.5	DIST. FROM SPINDLE CENTER TO COLUMN GUIDEWAYS	mm	510
2.6	COOLANT SUPPLY-FLOOD COOLANT AROUND THE SPINDLE	lpm	50
3	SPINDLE		
3.1	SPINDLE NOSE TAPER		7 / 24 No.40
3.2	SPINDLE SPEED - INDIRECT DRIVE	rpm	60 - 6000
3.3	SPINDLE BEARING DIAMETER	mm	70
4	FEED		
4.1	RAPID TRAVERSE X, Y & Z AXES	m / min.	30 / 30 / 30
4.2	CUTTING FEED	mm / min.	1 - 10000
4.3	GUIDEWAYS FOR X, Y & Z AXES		Linear Motion
5	AUTOMATIC TOOL CHANGER		
5.1	NUMBER OF TOOLS	Nos.	20
5.2	MAX. TOOL DIA	mm	80
5.3	MAX. TOOL DIA. WITH ADJACENT POCKETS EMPTY	mm	155
5.4	TOOL SELECTION		Bi - Directional
5.5	TOOL LENGTH MAX.	mm	250
5.6	TOOL WEIGHT MAX.	Kg.	8
5.7	CHIP TO CHIP TIME (WITH Z ONLY / WITH X & Y ALSO)	Sec.	3.9 / 4.8
5.8	TOOL CHANGE SYSTEM		Twin Arm
5.9	TOOL SHANK TYPE		BT - 40
5.10	PULL STUD TYPE		DIN 69872 Form-B
6	ACCURACY WITH ROTARY ENCODER		
6.1	POSITIONING FOR X, Y & Z AXES AS PER VDI DGQ 3441	mm	0.015
6.2	POSITIONING FOR X, Y & Z AXES AS PER ISO 230-2	mm	0.010
6.3	REPEATABILITY FOR X,Y&Z AXES AS PER VDI DGQ 3441	mm	± 0.005
6.4	REPEATABILITY FOR X, Y & Z AXES AS PER ISO 230-2	mm	± 0.003
7	MOTORS		
7.1	SPINDLE POWER 15 MIN/CONT. (FANUC)	kW	7.5 / 5.5
7.3	FEED MOTOR X, Y & Z AXES (FANUC)	Nm	11 / 11 / 20

ACE MANUFACTURING SYSTEMS LTD.

PRODUCT SPECIFICATION SHEET

AMS — In Pursuit of Excellence

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	1900
8.2	MACHINE DEPTH	mm	3150
8.3	MACHINE HEIGHT	mm	3250
8.4	MACHINE WEIGHT - NET	Kgf	4100
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	14
8.6	CNC CONTROL (WITH 8.4" COLOUR LCD)		FANUC 0IMATE-MD
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALL SCREW STRETCHING FOR X & Y AXES		
10.13	TOOL LIFE MANAGEMENT		
10.14	AIR DRIER UNIT (SMALL)		
11	**STANDARD CNC FEATURES OF FANUC-0IMATE-MD**		
11.1	THREE CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 3 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF THREE AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y CO-ORDINATES)		
11.15	TOOL OFFSET MEMORY - 400 NOS.		

ACE MANUFACTURING SYSTEMS LTD.			
PRODUCT SPECIFICATION SHEET			
MODEL: MCV-400			
Sl. No.	Description	Unit	MCV-400
11.16	TOOL LENGTH COMPENSATION		
11.17	PART PROGRAM MEMORY 1280 MTS (512K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
11.36	SPINDLE LOAD DISPLAY		
12	**MACHINE FEATURES / ACCESSORIES - ADDITIONALLY ORDERED BY M.S MATHARUENGINEERING WORKS, FARIDABAD.**		
12.1	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
12.2	CHIP FLUSHING SYSTEM ON THE BASE		
12.3	COOLANT GUN		

ACE MANUFACTURING SYSTEMS LTD

MACHINE SEQUENCE RUNNING TEST REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER : MIL-719

Preconditions for Machine before starting Machine Sequence Running

1. Machine should be fitted with all accessories as per Work Order ✓
2. Machine should be cleared from all Electrical & Mechanical Remarks ✓
3. Machine should be run with Sufficient Coolant. ✓
4. ATC should be full with tool holders ✓
5. No Bypass of any Safety switches and cabinet sealed. ✓

Date	Time	No of Hours Running	Date	Time	No of Hours Running	Remarks / Alarm Observed
28/9/M	5:00 PM	1 ✓			17 ✓	
		2 ✓			18 ✓	—
		3 ✓			19 ✓	
		4 ✓			20 ✓	
		5 ✓			21 ✓	
		6 ✓			22 ✓	
		7 ✓			23 ✓	
		8 ✓			24 ✓	
		9 ✓				
		10 ✓				
		11 ✓				
		12 ✓				
		13 ✓				
		14 ✓				
		15 ✓				
		16 ✓				

Nature of Remarks Related to	1.Hydraulic leakage	2. Coolant leakage	3.Electrical alarm	4. ATC	5. Spindle	6. Axes	7. Others

Note : In case M/c is stopped due to any reason, write the Sl. No. of the above remarks against corresponding number of hours, set right the M/c & continue to run for a total period of 24 hrs.

In charge - Mechanical **In charge - QA**

ACE MANUFACTURING SYSTEMS LTD

SPINDLE TEMPERATURE STABILIZATION REPORT

AMS — In Pursuit of Excellence

M/c. NUMBER : MH-~~432~~ 719	**MAXIMUM RPM :** 6000	**DRIVE: ID/DD/IDCTS/DDCTS**
SPINDLE SL No. : 10329	**PULLEY RATIO : 1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2**	

Note : Refer TM-QA-07 for test procedure. **Recommended Temp. Rise : 20°C max. above ambient**

% of current drawn at max RPM of the spindle : (Recommended max : DD : β3 = ≤ 15%, β6 / α6 & above = ≤10%. ID : β3 = ≤ 20%, β6 / α6 & above = ≤15%.	Measured = 10 %
Spindle noise at max RPM [Recommended : 80 dB(A) upto 10 kW, 85 dB(A) for > 10 kW]	Measured = 77 dB (A)

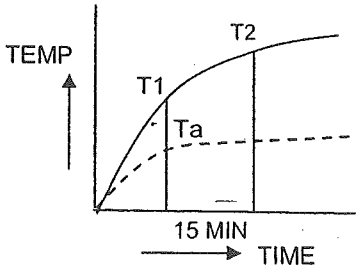
SPINDLE RPM	TIME	AMBIENT TEMPERATURE	SPINDLE TEMPERATURE
500	8:00	24°C	—
	9:00	25°C	30°
—	9:30	25°C	33°C
	10:00	25°C	35°C
	10:30	26°C	37°C
	11:00	26°C	37°C
	11:15	26°C	37°C
	11:30	26°C	37°C

1. The Stabilization point, Calculated as follows. Is the Temperature at which the rate of temperature rise over 15 minutes is less than 5%.

= [(T2 – T1) / (T1 –Ta)] X 100

= < 5%

T1 = Previous Temp., T2 = Temp. after 15 Min & Ta = Ambient Temp. at T1



2. RUN SPINDLE AFTER STABILIZATION FOR 3 HOURS.

Date 25/9/11

Checked By

F-QA-FA-11-R0

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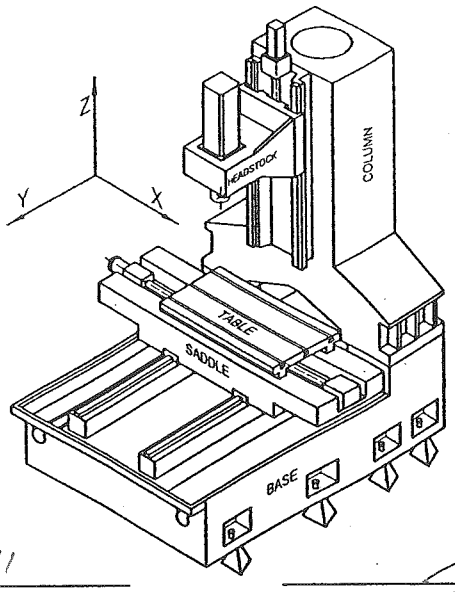
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

PRODUCT : VERTICAL MACHINING CENTER (FIXED COLUMN)

MODEL NAME : MCV-400

MACHINE NUMBER : MCL - 719



28/9/11

CHECKED BY
QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

ACE MANUFACTURING SYSTEMS LTD

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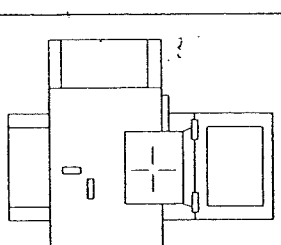
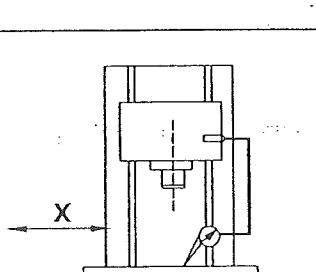
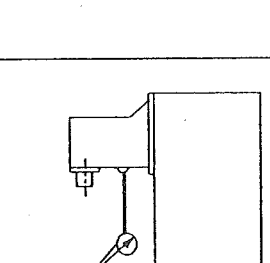
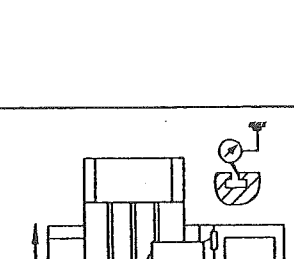
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / m ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0.008 LEVEL ALONG Y-AXIS : 0.005
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.008 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.005 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.006 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)

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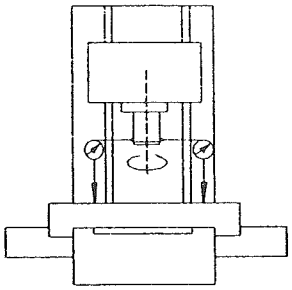
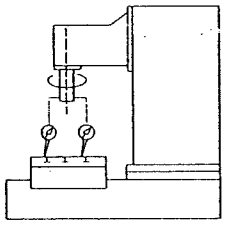
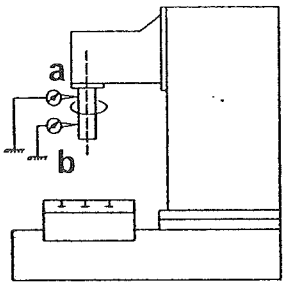
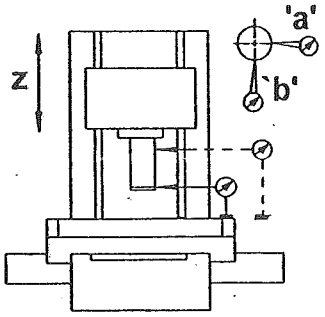
DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.002 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G6	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.012. PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR SPINDLE NOSE b) AT A DISTANCE OF 300 FROM SPINDLE NOSE		a. 0.01 NEAR SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 0.002. b. 0.014
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 0.004 b. 0.007

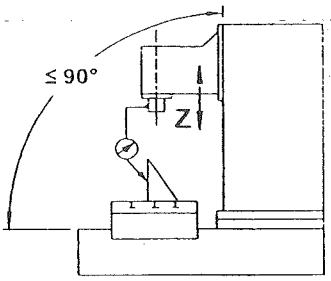
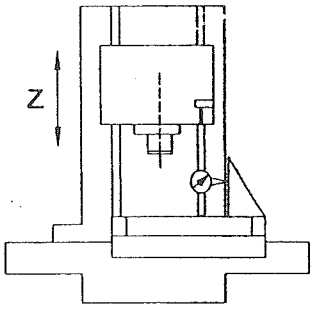
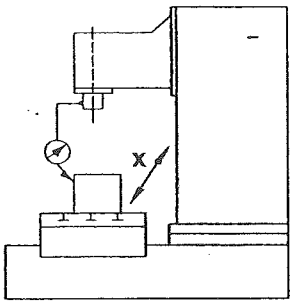
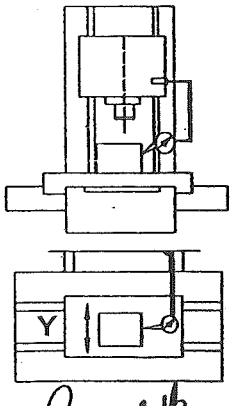
SIGN & DATE	BSR	IG	RP	31.08.11	3 OF 5	DS-DSN-TC-PROD-01-R2
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

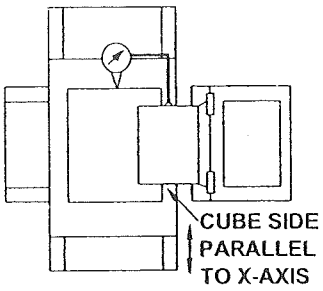
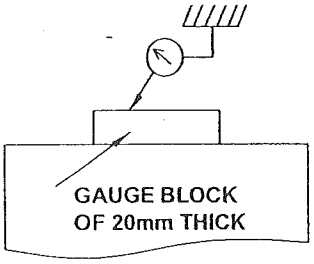
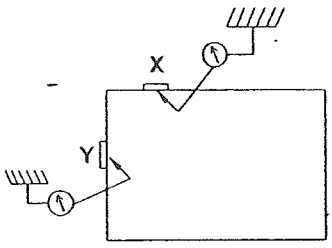
SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN YZ PLANE		0.015 / 300	TABLE : 0.008 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN XZ PLANE		0.015 / 300	TABLE : 0.010 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINING CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G13	SQUARENESS OF Y - AXIS MOVEMENT WITH RESPECT TO X - AXIS MOVEMENT	MEASURE READING ALONG 'Y' 	0.015 / 300	0-005
G14	VARIATION BETWEEN THE PALLET HEIGHT (FOR APC MACHINE ONLY)		0.030	NOA
G15	PALLET REPEATABILITY IN X & Y AXES WHEN LOADED ON THE MACHINE FROM THE PALLET STATION FOR SAME PALLET (FOR APC MACHINE ONLY)		0.020 IN X & Y AXES DIRECTION	PALLET 1 : NOA PALLET 2 : (FOR APC MACHINE ONLY)

SIGN & DATE	BSR	IG	RP	31.08.11	5 OF 5	DS-DSN-TC-PROD-01-R2
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

AMS — In Pursuit of Excellence

1) HEAVY CUT REPORT :

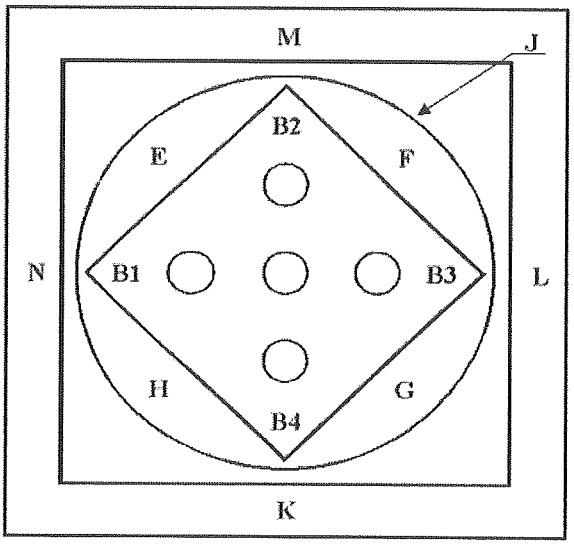
MODEL	: MCV-400	BASE SPEED OF MOTOR	:1500 rpm	MATERIAL	: C.I.(Gr 25)
M/C NO.	: M11-719	SPINDLE SPEED RANGE	:60-6000 rpm	CUTTER	: Ø80 Sq.Sh FMC
TYPE OF MOTOR	: βi 6/10000	MOTOR SPEED RANGE	:90-9000 rpm	WIDTH OF CUT	: 55 mm
PULLEY RATIO	: 1:1.5	MOTOR CAPACITY	:7.5/5.5 kW	C.S. (V)	: 250 mtr/min.

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as per CRT	Spl.load % as per CRT	Remarks
1000	0.15	1.05	1050	2.0	116	28	58	SMOOTH CUTTING
1000	0.15	1.05	1050	3.9	225	46	107	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NAS TEST REPORT :

No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings in mm
1	Flatness of Surface	A	0.015	**0.008**
2	Parallelism of Surfaces	N-L	0.020	**0.006**
		M-K	0.020	**0.004**
		H-F	0.030	**0.006**
		G-E	0.030	**0.009**
3	Squareness of side faces	K-L	0.020	**0.007**
		L-M	0.020	**0.003**
		M-N	0.020	**0.006**
		N-K	0.020	**0.008**
		E-F	0.030	**0.009**
		F-G	0.030	**0.004**
		G-H	0.030	**0.009**
		H-E	0.030	**0.003**
4	Circularity of Machining	J	**0.040**	**0.015**
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	**-0.001**
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	**-0.003**
	Diagonal pitch	C(70.71)	0.035	**-0.002**
6	Bore size variation		0.020	
	Hole No.	**Max.**	**Min.**	**Ovality**
	B1=	26.558	26.554	**0.004**
	B2=	26.557	26.555	**0.002**
	B3=	26.559	26.554	**0.005**
	B4=	26.555	26.552	**0.003**
7	Straightness	K	0.010	**0.005**
		L	0.010	**0.004**
		M	0.010	**0.002**
		N	0.010	**0.005**
		E	0.015	**0.005**
		F	0.015	**0.004**
		G	0.015	**0.006**
		H	0.015	**0.002**
8	Step difference in axis change over		0.010	**0.004**



SINGLE AXIS CONTROL K=L=M=N = 240 mm

DOUBLE AXIS CONTROL E=F=G=H = 160 mm

CIRCLE " J " = Ø 234.70mm

Heavy cut Done By	NAS TEST DONE BY	Department incharge	QC / CMM
For BNK	MANJU		
Date: 24/09/2011	DATE:27/09/2011	Date:	Date: 29/9/11

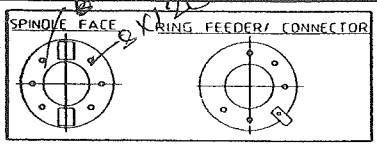
ACE MANUFACTURING SYSTEMS LTD

SPINDLE BALANCING REPORT

AMS — In Pursuit of Excellence

Machine Number :	MM-719
Spindle Serial Number :	10132
Spindle Type :	CDSTD
Spindle Speed (Max.) :	6000 rpm.

Note : Refer TM-QA-09 for measurement procedure.



Location of balancing screw(s)

Sl No.	Speed in RPM	Velocity of vibration (mm/sec) before Balancing Along X-axis	After Balancing along X-axis (8000 rpm : 0.8 mm/sec max. 8000 rpm : 1 mm/sec max.) >	After Balancing along Y-axis (8000 rpm : 0.8 mm/sec max. 8000 rpm : 1 mm/sec max.) >
1	500			
2	1000	0·021	0·005	0·001
3	1500	0·015	0·002	
4	2000	0·020	0·005	0·004
5	2500	0·077	0·012	
6	3000	0·143	0·012	0·009
7	3500	0·223	0·034	
8	4000	0·350	0·050	0·048
9	4500	1·21 – 125°	0·104	
10	5000	2·63 – 38°	0·245	0·215
11	5500	1·08 – 21°	0·168	
12	6000	1·35 – 8°	0·175	0·056
13	6500			
14	7000			
15	7500			
16	8000			
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle maximum vibration mm / sec : | **Correction in gms :**

REMARK IF ANY:

Date: 29/09/14 | Checked By | APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

SPINDLE ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

Machine Number : MH-719	Spindle Unit Number: 10132
Spindle Drive ID / ID -CTS / DD / DD-CTS	Max. Spindle Speed: 6000 RPM
Pull stud type DIN / MAS	Draw bar Type ID / ID-CTS / DD / DD-CTS
Bearing Make & Specification : FAG B7014 ETP4/DU	Draw bar SL No: 0860
Spindle Number: 1479	Cartridge Number: 611
Spindle Nose Taper ISO-30 / 40 / 50 / HSK50 / 63 / 100	

SL No.	Description	Recommended	Actual
1	Locknut Tightening Torque	Ref.DS-PDN-MEC-SA-08	118 Nm
2	Spindle Bearing Preload	Ref.DS-PDN-MEC-SA-07	-0.009 mm
3	Spindle Runout on taper	0.002 mm (*)	0.002 mm
4	Mandrel Runout near the spindle nose	0.005 mm (*)	0.002 mm
5	Mandrel Runout at 4 positions @ 90° at a distance of 300 mm (ISO 40/50, HSK 63), 250 mm (ISO 30, HSK 50) from spindle nose	10 micron (*)	1. 4 2. 3 3. 4 4. 4
6	Spindle stabilization temperature on Test rig	Upto 15° C above ambient	31°C - 31°C = 10°C
7	Check for abnormal / unhealthy noise	Abnormal / unhealthy noise not acceptable	ok
8	Pulley ratio (For ID machines)	Refer Work order	1:1.5 ratio
9	Number of Teeth on spindle pulley (For ID machines)	Ref.DS-DSN-PROD-02	81 teeth
10	Spindle pulley Runout (For ID machines)	0.050 mm (*)	0.050 mm
11	Spindle connector Runout (For DD machines)	0.010 mm (*)	—
12	Disc spring specification (Single / double stack)	Ref.DS-DSN-PROD-01	Ø100 x Ø0.8 x 3 mm
13	Number of disc springs used	Ref.DS-DSN-PROD-01	66 Nos
14	Disc spring Compression, Free length & Compressed length	Ref.DS-DSN-PROD-01	272 - 251 = 21 mm
15	Tool clamping force	Ref.DS-DSN-PROD-01	8 kN
16	ID-CTS Deublin Runout in Assembly condition (For CTS Spindle)	0.030 mm	Deublin Sl. No. : ..... Run out : .....

(*) To be measured after temperature stabilisation of spindle at max. speed.

REMARK IF ANY:

Date: 14/9/2011	Checked By	Approved By

F-QA-IPR-SPL-01-R0

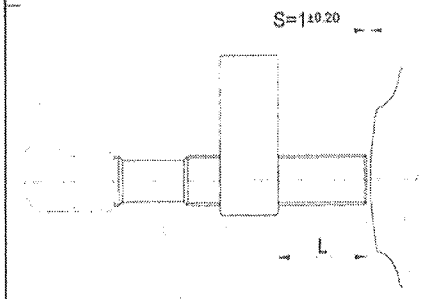
PRAGATI	AUTOMATIC TOOL CHANGER TEST REPORT	
SL NO: 13839	CUSTOMER I.D: SOL-011-78	CUSTOMER: AMS
PRODUCT I.D.: ATC4020V-029	PRODUCT DESCRIPTION: ATC4020V-R-225-BT-MAS	
DESPATCH DATE: 06-09-2011		

TEST RESULTS:

TESTING HOURS	SPECIFIED: 6 Hrs.	ACTUAL: 6 HOURS

PARAMETERS	SPECIFIED	ACTUAL		
		DURING ASSLY	AFTER 6 Hrs	DURING DESPATCH
ARM STROKE	114 -115mm	114.00	114.30	114.30
ARM AXIAL PLAY(IN PARKING)	0.20-1.50mm	1.00	1.20	1.20
ARM RADIAL PLAY @R250 IN PARKING	0.20-1.0mm	0.30	0.40	0.40
ARM RADIAL PLAY @R250 WHILE MOVING OUT	0.20-1.0mm	0.30	0.50	0.50
ARM RADIAL PLAY @R250 WHILE MOVING IN	0.20-1.0mm	0.30	0.60	0.60
ARM PARLLALITY	0-0.20mm	0.10	0.20	0.20
ARM PARK SIGNAL	Visual Check	OK	OK	OK
TOOL DECLAMP SIGNAL	Visual Check	OK	OK	OK
TOOL CLAMP SIGNAL	Visual Check	OK	OK	OK
BACKLASH OF WHEEL @R300	0.05-0.40mm	0.15	0.20	0.20
AXIAL PLAY OF WHEEL	0.01-0.10mm	0.08		

SENSING DISTANCE



PROXIMITY	SENSING GAP(S)	SENSOR LENGTH(L)
PS1	1.00	22.90
PS2	1.00	23.00
PS3	1.00	23.20
PS4	1.00	30.00
PS5	1.00	—
PS6	1.00	—
PS7	1.00	—

DESCRIPTION	NAME	SIGNATURE	DATE
ASSEMBLED BY	Manjunatha.PG	Manju	04/09/11
CHECKED BY AFTER 6Hrs	Hanumappa	HSP	05/09/11
INSPECTED BY	Ayyappa.C	Ayyappa	06-09-11
APPROVED BY	Chamalaraya	Chamalaraya	06-09-11

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

MACHINE NUMBER: MH-719

I. Lubrication metering cartridge specification in CC:

Sl. NO	Axis	Ball Screw	Guideway Top / LM Block	Guideway Side	Jib	Keeper plate	
1	X-Axis	0.25cc	0.1cc				
2	Y-Axis	0.25cc	0.1cc		N/A		
3	Z-Axis	0.25cc	0.1 cc				

Sl. NO	Check points:	Remark
1	Ensure X,Y & Z guideways and ball screw are lubricating properly	Ok
2	Ensure lubrication pressure set to 15 Bar.	Ok
3	Ensure there is no leakage of lubrication oil from joints, cartridges & manifolds etc.	Ok
4	Ensure that the lubrication hoses do not rub or foul with the stationary parts during axis movement.	Ok
5	Ensure that the lubrication hoses of appropriate length are used and properly clamped and routed.	Ok
6	Ensure metering cartridge are fixed as per circuit diagram.	Ok

II Telescopic cover Assembly:

	Sl No	Axis	Extension block Parallelism		Length Variation -	Telescopic Cover
			Top -Permissible 0.020 mm	Side-Permissible 0.030 mm	Permissible 0.050 mm	drawing Number
	1	X - Motor End	0.010	0.020	0.040	
	2	X - Brake End	0.010	0.010	0.040	
	3	Y - Motor End	0.020	0.010	N/A	
	4	Y - Brake End	0.010	0.010	0.030	
	5	Z - Brake End	0.020	0.010	0.030	

III. Saddle cover alignment (If required):

	Sl NO	Top - Permissible 0.050 "mm"	Side - Permissible 0.050 "mm"	
	1	N/A	N/A	

Sl NO	Check points:	Remark
1	Check telescopic cover sliding smoothly	Ok
2	Ensure Top & Side surface of the extension blocks are matching with respect to guideways.	Ok
3	Ensure telescopic cover screws are tightened in box close condition	Ok

30/09/14	C.S. Mehta		30/09
Date:	PREPARED	CHECKED	APPROVED

ACE MANUFACTURING SYSTEMS LTD

LUBRICATION ASSEMBLY - INSPECTION REPORT

MACHINE NUMBER : MIL-719

IV. Motor Assembly:

Motor mounting bracket Concentricity & Face out:

Sl No	Axis	Face out -Permissible 0.030 mm	Concentricity-Permissible 0.060 mm
1	X-Axis	0.020	0.050
2	Y-Axis	0.020	0.040
3	Z-Axis	0.040	0.050

Sl NO	Check points:	Remark
1	Ensure motor shaft and coupling bore are free from oil / grease before fitting of coupling	Ok
2	Ensure Axis Motor & Motor mounting bracket are fitted as per specification.	Ok
3	Ensure coupling 3 slots (Steel Bellows, Clamping ring & Bush) are in same line.	Ok
4	Check the tightness of coupling and set to 17 N-m (M6 screw), 35 N-m (M8 screw) Torque.	Ok

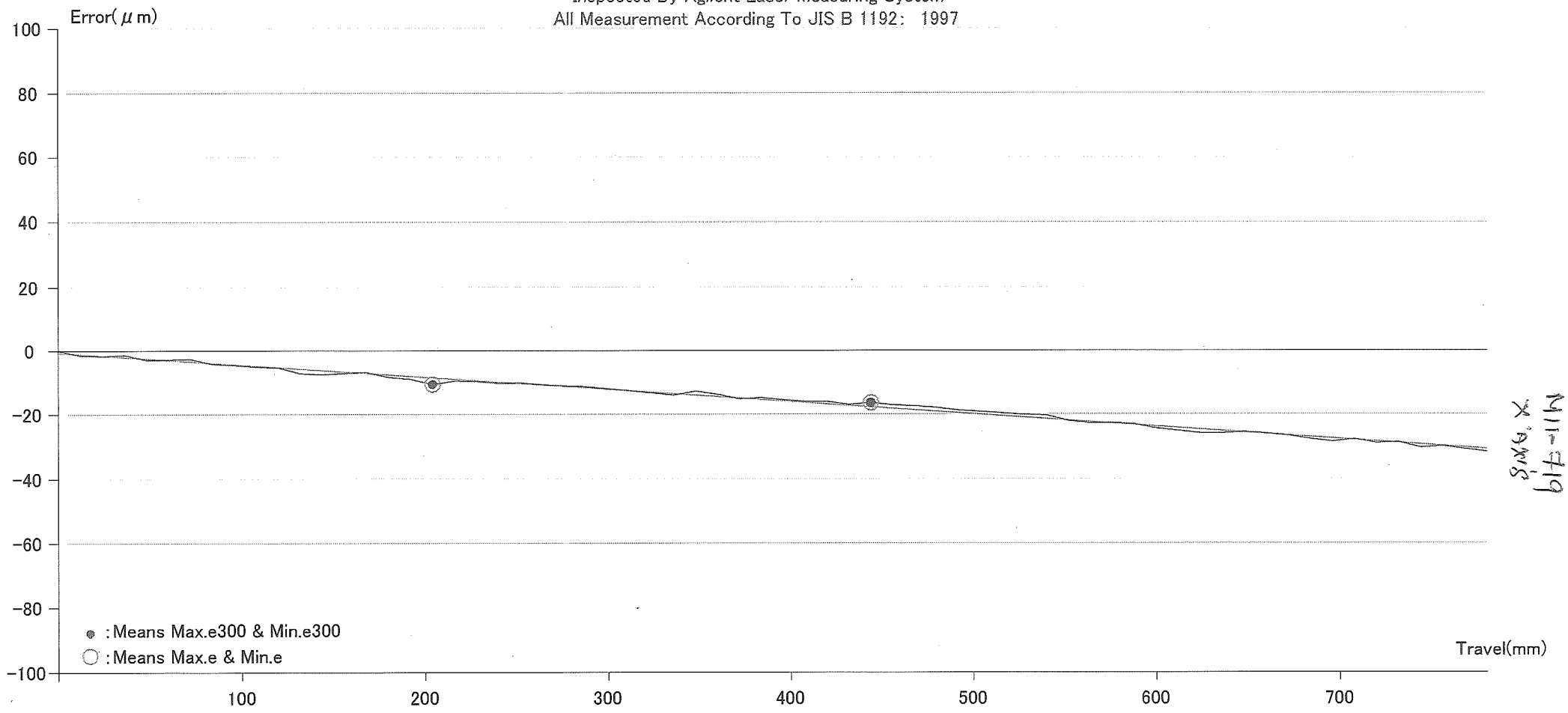
V. Counter balance Assembly (If Required)

Sl NO	Check points:	Remark
1	Ensure Smooth movement of Z- Axis with out fouling of dead weight (MCV 500 / MCV 600 only)	N/A
2	Ensure the parallelism of counter balance cylinder piston is with in 0.050 mm both XZ & YZ plane with respect to Z Axis movement.	N/A
3	Ensure Slide is balanced properly in both upward and down ward movement	N/A
4	Ensure the counter balance bracket Dowelling.	N/A

19/09/19	C.S. Mehta		30/09/19
Date:	PREPARED	CHECKED	APPROVED

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-110001331002
·Drawing No.	:BA300510002-0
·Part No.	: *:S12-016-95
·Specification	:R40-12C1-FDWE-846-1110-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43～-7	-30.15
·e(μm)	13	3.45
·e300(μm)	8	3.45
·Tq(kgf - cm)	4.16～7.28	5.2～6.24
·Accuracy Grade	C3	C0

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/06/06
PASS

Inspector: Angus

ACE MANUFACTURING SYSTEMS LTD

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BALL SCREW ASSEMBLY REPORT

Machine Number : M11-#19 **Ball screw Sl No:** 11000131002 **Axis :** X

I. Ball screw Make & Specification : PMI (R40-10(1)-TDWE-846-1110-0.008)

II. Ball Scerw Bearing Make & Specification : NACHI (25TAB06DB(UN)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.2N/M	1.3N/M
2	Bearing housing Bore diameter	As per drawing	6to3u	7to8u
3	Bearing housing centre height difference	0.01 mm	0.01MM	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.008MM	0.006MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.930
4	Bearing width B3	—11—	T4	14.930
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	(2U-275-8128	T6	—
7	Housing Bearing depth	(37-103-2123	T7	43.480

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103/11

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.930
4	Bearing width B3	—11—	T4	14.930
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	(2U-360-8128	T6	—
7	Housing Bearing depth	(37-153-2128	T7	43.480

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153/11

T6 = 13.680 mm

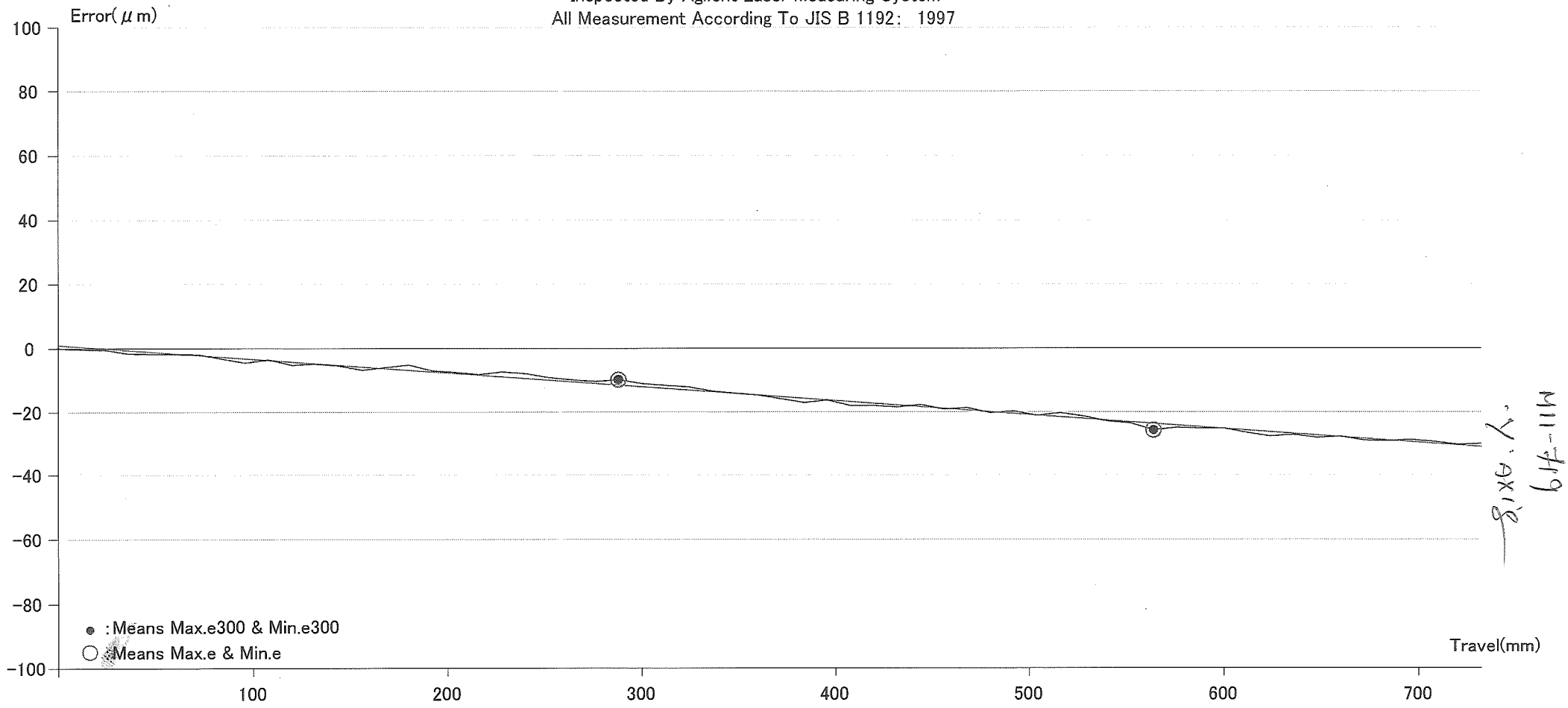
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	—11—
3	Ensure smooth rotation of ball screw assembly.	—11—
4	Ensure ball screw run out within 0.010 mm	—11—
5	Ensure tightness of all screws & marking with yellow marker	—11—
6	Ensure ball Screw report attached to Inspection Report.	—11—

Date: Checked by Approved by

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



M11-479

Specifications:	
·Customer	:AMS
·Order No.	:S3-110007463002
·Drawing No.	:BA300510003-0
·Part No.	: *:S12-016-97
·Specification	:R40-12C1-FDWE-796-1060-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:732
·E(μm)	:±18
·T(μm)	:-24

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-42～-6	-31.96
·e(μm)	13	3.77
·e300(μm)	8	3.77
·Tq(kgf－cm)	4.16～7.28	5.2-6.24
·Accuracy Grade	C3	C1

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/6/21
PASS

Inspector: LUIGI

ACE MANUFACTURING SYSTEMS LTD

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BALL SCREW ASSEMBLY REPORT

Machine Number : M11-719 | Ball screw Sl No: 1000746 3002 | Axis : Y

I. Ball screw Make & Specification : PMI/R40-12C4-FDWC-796.1060-0.008)

II. Ball Screw Bearing Make & Specification : NACHI (25TAB06DB/UM)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.2N/M	1.3N/M
2	Bearing housing Bore diameter	As per drawing	4to32	4to32
3	Bearing housing centre height difference	0.01 mm	0.01MM	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.008MM	0.006MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.935
4	Bearing width B3	〃	T4	14.935
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C20-275-8123	T6	—
7	Housing Bearing depth	C37-103-2128	T7	43.490

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103/19

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.945
4	Bearing width B3	〃	T4	14.945
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C20-360-8123	T6	—
7	Housing Bearing depth	C37-153-2123	T7	43.510

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153/19

T6 = 13.680 mm

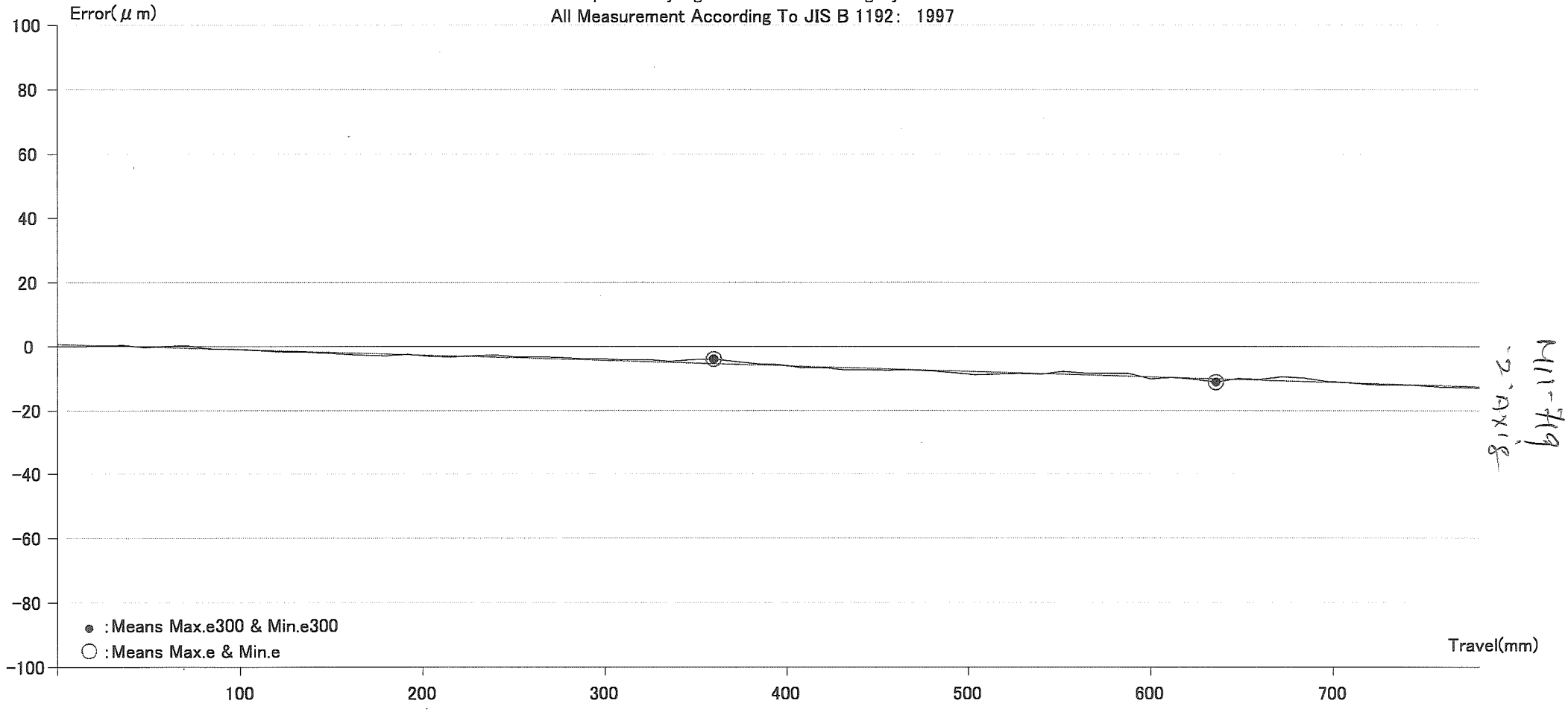
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	〃
3	Ensure smooth rotation of ball screw assembly.	〃
4	Ensure ball screw run out within 0.010 mm	〃
5	Ensure tightness of all screws & marking with yellow marker	〃
6	Ensure ball Screw report attached to Inspection Report.	〃

Date:	Checked by	Approved by

PMI BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-110001313012
·Drawing No.	:BA300510008-0
·Part No.	: *: S12-016-114-Z(端塞)
·Specification	:R40-12T5-FDDC-841-1105-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:0

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	−18～18	−13.34
·e(μm)	13	2.48
·e300(μm)	8	2.48
·Tq(kgf－cm)	5.70～10.59	6.2ψ7.28
·Accuracy Grade	C3	C3

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 0.5℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

QC
2011/06/20
PASS

Inspector: ARJAY

ACE MANUFACTURING SYSTEMS LTD

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BALL SCREW ASSEMBLY REPORT

Machine Number : M11-719 **Ball screw Sl No:** 110001313012 **Axis :** 2

I. Ball screw Make & Specification : PMI/R40-12TG-TDDC-845-1705-0.008)

II. Ball Screw Bearing Make & Specification : NACHI/25TAB06DB(UM).

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.2N/M	1.3N/M
2	Bearing housing Bore diameter	As per drawing	4 to 14	9 to 69
3	Bearing housing centre height difference	0.01 mm	~~0.01 MM~~	-
4	Bearing housing bore parallelism with respect to pads	0.01 mm	—	0.006MM

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	MU-291-8123	T1	13.015
2	Bearing width B1	-	T2	-
3	Bearing width B2	NACHI	T3	11.941
4	Bearing width B3	—11—	T4	14.955
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	MU-291-8123	T6	-
7	Housing Bearing depth	137-701-2123	T7	56.987

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 701/29

T6 = 184.155 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	-	T1	-
2	Bearing width B1	-	T2	-
3	Bearing width B2	630513	T3	-
4	Bearing width B3	-	T4	-
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	MU-280-8123	T6	-
7	Housing Bearing depth	137-150-2123	T7	43.460

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 154/45

T6 = mm

VI. Check points after assembly

	CHECK POINTS	Remark
1.	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	—11—
3	Ensure smooth rotation of ball screw assembly.	—11—
4	Ensure ball screw run out within 0.010 mm	—11—
5	Ensure tightness of all screws & marking with yellow marker	—11—
6	Ensure ball Screw report attached to Inspection Report.	—11—

Date:	Checked by	Approved by

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MACHINE HEALTH CHECK POINTS

MACHINE NUMBER : MC-779 **DATE :** 28/9/11

I	**AXIS**	
1	Ensure plugs are fitted for L M Guide ways.	—
2	Ensure effective wiping of saddle cover wipers.	T. Cover.
3	Ensure the lube timer is set 12 min for HG and 20 min for LM Guide ways.	—
4	Ensure leveling bolts are fixed with locknut.	✓
5	Check the Table tenon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm)	✓
6	Ensure butt reference function.	✓
7	Check all the axes reference arrows are matched.	✓
8	Check Emergency push button function of Machine / APC / Double Hand	✓
9	Ensure marking of all screws with Yellow paint marker	✓
10	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	NA
11	Ensure smooth movement of sliding door without play and unhealthy noise	✓
II	**SPINDLE**	
1	Ensure air jet through all the holes (4/8 holes) at oriented spindle position	✓
2	Ensure supply of air to spindle Air Purge system	NA
3	Check the tightness of push stud.	✓
4	Ensure grub screws are fitted to spindle flange.	✓
5	Check spindle orientation position and note down the value. (Parameter No. : 4031 for Fanuc controller, ... for Siemens controller)	860 ✓
6	Ensure spindle balancing is completed	✓
7	Ensure presence of gap between draw bar bush & ejection screw / roller (visually or by feel)	✓
8	Check the "Tool not in spindle" proximity message.	NA
III	**ATC**	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Ensure oil level is max. in ATC gear box.	✓
3	Check the Tool Number display on panel board is same as in spindle	✓
4	Check tool change position limit switch for proper functioning & band width (Max. 1.5 mm on each side).	✓

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ACE MANUFACTURING SYSTEMS LTD MACHINE HEALTH CHECK POINTS		AMS In Pursuit of Excellence
IV		COOLANT
1	✓	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe
2		Ensure proper working of float switch (by shifting the sensor upwards). ✓
3	✓	Check the function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).
4		Ensure drum wash in conveyors (Optional)
5		Check conveyor overload message & Direction (Optional)
6		Check oil in conveyor gear box. (Optional)
7		Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).
8		Ensure CTS flow switch setting by opening the bag filter air vent ball valve (Optional)
9		Ensure suction strainer & Check Valve in CTS tank & pr. gauge functioning (Optional)
10		Ensure CTS pin locator hole is plugged (Optional)
11		Ensure that twin arm plunger & CTS pin locator screw do not foul (Optional)
12		Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)
13		Ensure High pressure air blast after CTS switched off (Optional)
V		APC
1		Check pallet initialization in APC Machine
2		Ensure supply of air jet to pallet rest pad cleaning.
3		Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise
VI		HYDRAULIC, PNEUMATIC & LUBRICATION
1		Ensure air pressure switch is working properly.
2		Ensure proper setting of oil (1 drop / tool change) in FRL unit & Oil Level low message
3	✓	Check the FRL drain hose is fixed.
4	NA	Ensure identification name plates are fixed on power pack hoses.
5	NA	Check the Hyd.oil level low message (by adjusting sensor position) & Pr. gauge function
6	NA	Ensure hydraulic flow control valve grub locking in APC machine.
7	✓	Ensure Ball valve fixed to FRL Unit inlet
8	NA	Ensure no oil leakage in hydraulic valves & fittings
9	✓	Ensure lubrication pressure switch is set to 15 bar.
10	NA	Ensure pneumatic quick change coupling fixed to 4th axis provision
VII		GENERAL
1	NA	Check all the points of MOM are attended.
2		Ensure coolant tank capacity engraved or marked in name plate
3	NA	Check Hydraulic specification is as per circuit diagram
4	✓	Ensure axis calibration & ball bar parameter as per the report.
5	✓	Ensure Ball screw pitch in the machine and report is same.
6	✓	Ensure spindle Sl. No. and parameters are as per the report.

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ACE MANUFACTURING SYSTEMS LTD.

AMS — In Pursuit of Excellence

MACHINE HEALTH CHECK POINTS

MACHINE No. : MLL-719 ASSEMBLY TECHNICIAN : Bheema Rao & Team.

S.No	REMARKS OBSERVED	STATUS	SIGN
(1)	RH Side headstock guard foling to the limite switch bracket	✓	✓
(2)	Coolent Level float Switch fixing		
(3)	Lubrication oil leakage from cenduble unite banjo	✓	✓
(4)	FRL oil Leting to be done	✓	✓
(5)	Single pendant Stopper fixing to be done.	✓	✓
(6)	Rear Side of the M/c cable rooting property (E)		✓
(7)	Base build ball screw, spindle, spindle balancing reports. to be filed		
(8)	Turn around reading in y direction Measures 0-03mm.	✓	✓

ASSEMBLY IN CHARGE

CHECKED BY

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

CHECK LIST - MECHANICAL

MACHINE NUMBER : MU-719 **DATE :** 28/9/11

I	Z-axis motor current during tool declamp (Turret type ATC) : Recommended Max : BT30-150%, BT40-180%)	
A	With Z axis moved at 840 mm/min in Jog mode	NA
B	In Auto cycle	

II Z - axis drop during power off :

- ve' direction< 0.005 mm		'+ ve' <0.015mm direction	
System Initial Reading :	250.000	System Initial Reading. :	250.000
System Final Reading :	249.998	System Final Reading. :	249.984
System Diff. in Reading :	+0.002	System Diff. in Reading. :	0.016
Dial Reading. :	+0.002	Dial Reading. :	0.006

III % of Axis current drawn (Recommended as per Data sheet DS -QA -02)

A	Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm/m Feed rate
	X	12%	17%	48%	12%
	Y	18%	23%	51%	19%
	Z Up	56%	58%	65%	53%
	Z Down	32%	28%	29%	34%
	B - Forward				
	B - Reverse				

B % of Axis holding current (Recommended as per Data sheet DS -QA -02)

Axis	100% Rapid Rate		2000 mm/min. Feed Rate	
	Motor end	Brake end	Motor end	Brake end
X	9%	10%	2%	3%
Y	11%	12%	2%	1%
Z	54%	36%	44%	44%

C Over travel against Axis dead stopper in mm (Recommended as per Drawing)

Axis	Brake end	Motor End
X	21.095	25.000
Y	17.392	21.107
Z	9.692	21.261

IV SPINDLE

A	Push stud stroke length for Tool ejection (mm)	Recommended values	Auto	Manual
	BT – 30/40 /50	Refer DS-DSN-PROD-01	0.800	0.800
	HSK A50/63/100		NA	NA
B	Tool clamping force-Kgf	Refer DS-DSN-PROD-01	810 Kgf	

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C	Axial play in the spindle (Ref. TM-QA-10 for procedure) (Recommended max.except BT50 : 0.005 mm , for BT 50 : 0.008 mm)		
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ + 0.030 ⟶ 0.000	Difference = mm (R1~R2) 0.002
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ - 0.024 ⟶ 0.022	
V	Turret type ATC		
A	Check the spring lever compression at :		
	1. After the magazine butts against the dead stopper (mm)		
	2. Z – axis position below zero position (mm)		
	3. Recommended : BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm (i.e. 1-2)		
B	Tool picking position for Turret Type ATC Recommended : BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm		
VI	A.P.C		
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram)		
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram)		
VII	Cycle Time in seconds (Refer Work order for recommended value)		
	1. Check the tool to tool time		2.0 Min
	2. Check the chip to chip time with Z axis only		3.95 Sec
	3. Check the chip to chip time with all axes		4.850 Sec
	4. Check the pallet changing time		WA

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CHECK LIST – MECHANICAL

AMS

MH-719

1. MACHINE LEVEL READING :

Along X-Axis			Along Y-Axis		
½	½	½	½	½	½
½	0	½	½	½	½
½	0	½	½	½	0

2. TABLE TOP READINGS:

Along X-Axis		
0	-4	-7
0	-4	-7
0	+2	+3

Along Y-Axis		
0	0	0
-2	0	+2
-5	-2	+3

3. TENON SLOT READING :

0	+6	0

PALLET – II

NA

PALLET – II

Along X-Axis	Along Y-Axis
NA	NA

5. MANDREL READINGS:

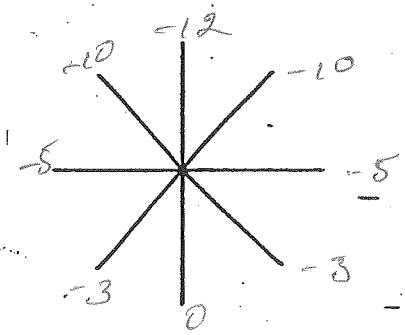
Front	Side
+1	+4
0	0

RUNOUT:

a. TOP : (0.010mm)
1. 0.002

b. BOTTOM :(0.020mm)
1. 0.008
2. 0.005
3. 0.012
4. 0.014

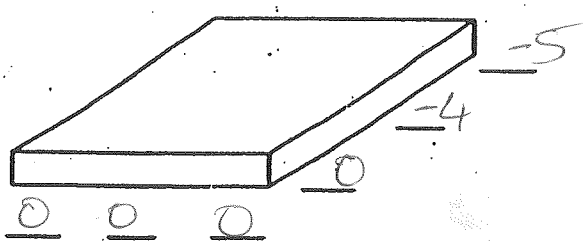
6. TURN AROUND READING:



4.COLUMN READINGS :

FRONT	SIDE
-10	-8
0	0

7. SQUARENESS READING (X & Y):



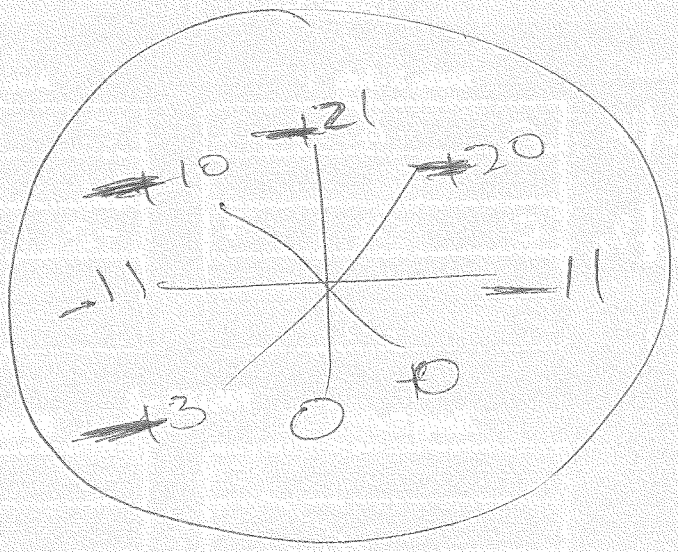
8. PALLET HEIGHT VARIATION:

Pallet-1: NA

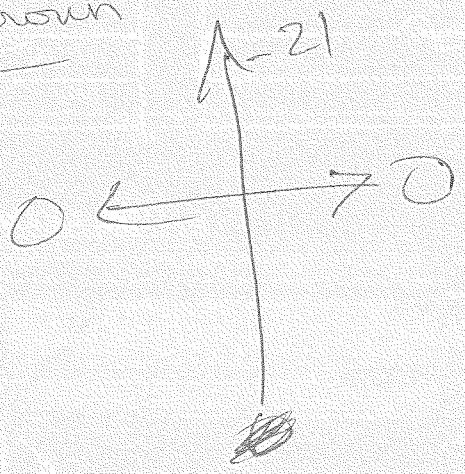
Pallet-2 : _____

NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01-06.

CHECKED BY



Turn Around



Tenroom

-20 ← 0

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - MECHANICAL

AMS — In Pursuit of Excellence

MACHINE NUMBER : MH-719 **DATE :** 28/9/11

1	ATC Details : Model	4020V / 4024V / 4030V / 4020H / 4024H /HSK A63 /HSK A100		
1.1	ATC :Turret Type/Twin Arm	Sl.No	13839	
1.2	Turret type magazine Encoder	Make	Model	Sl. No
		Pragati	ATC12 / ATC16 / ATC20	
1.3	ATC Torque Motor	TM 50 / 63 / 80 / 100 / 125 / Servo ATC		
2	**BELT / COUPLING DETAILS :**			
		GT Belt (For Indirect Drive)	COUPLING (For Direct Drive)	
2.1	Make	Unitta / Power Grip	Mayr	
2.2	Model			
3	**COOLANT SYSTEM DETAILS :**			
	Make : Rajmane Industries		Model	Type
3.1	Flood Coolant Pump - Crown/Flexinox/Disc Nozzle		RKM 206 / RV 160 - 270	Centrifugal Pump
3.2	Chip Pushing Coolant Pump		RKM 206 / RV 160 - 270	
3.3	Base Flushing Coolant Pump		RKM 206 / RV 160 - 270	
3.4	Coolant Gun & Drum Wash Pump		RKM 206 / RV 160 - 270	
3.5	Oil Skimmer - Make / Model			
3.6	High Pressure Coolant Pump		Make	GF/ WP / BP
			Model	
3.7	Pressure Relief Valve		Make	Polyhydron
			Model	DPRH-10T-100-04
3.8	Rotary Joint		Make	Deublin / RIX
			Model	
4	**FLOAT SWITCH :**			
	Make		Filpro Electronics	
5	**CHIP CONVEYOR DETAILS :**			
5.1	Type of Conveyor	Scraper / Slat cum Scraper / Magnetic along X or Y axis.		
5.2	Model			
5.3	Make	Universal Conveyor		
5.4	Sl.No			
6	**CONVEYOR GEAR BOX DETAILS :**			
6.1	Model			
6.2	Make	Universal Conveyor		
7	**FILTER SPECIFICATIONS :**			
7.1	Model / Working pressure	8 /15, 50 bar / 100 bar		
7.2	Make	Universal Conveyor		
7.3	Sl.No			
8	**COIL / SCREW CONVEYOR DETAILS :**	Qty : 1 / 2		
8.1	Make	Universal Conveyor		
8.2	Model			

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ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - MECHANICAL

AMS — In Pursuit of Excellence

9	HYDRAULIC SYSTEM :			
9.1	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi /Hymech	
9.2	Product No		QC. No :	
9.3	Tank Capacity (Litres)		32 / 40 / 63 / 70	
9.4	Maximum Working Pressure			
9.5	HYDRAULIC PUMP	Make Rexroth / yuken /Corotech / vickers	Model	
9.6	MOTOR	Make Rotomotive / Siemens	Model 90L-4-1400 / 90S-4-1390 ,	
10	PRESSURE SWITCHES FOR HYDRAULIC SYSTEM :			
10.1	Make		Polyhydron	
10.2	Model		1PS 30-50 / 30-100 / 10 - 30 /	
10.3	Quantity			
11	HYDRAULIC ACCUMULATOR :			
11.1	Make	FOX / BOSCH	FOX / BOSCH	
11.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
11.3	Gas Pressure	Tool de clamp :	Counter balance:	
11.4	Volume (Litres)	Tool de clamp : 1.4	Counter balance : 4 or 10	
12	HYDRAULIC CYLINDERS :			
	Make	Model	Function	Qty
12.1	Island Hydraulics	Ø 40 X 22 X [125/152/180/1150]	Pallet Forward / Reverse	
13	LUBRICATION UNIT DETAILS	Make	Pressure Setting	
		CENLUB / DROPCO	15 Kg/ cm2	
14	FRL UNIT			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan	MINI / MIDI	0-16 Kg/cm2	Servo Way Spin 12
15	DRYER	Make		Specification
		Trident		VXD - 2
16	PNEUMATIC VALVES			
	Application	Specs	Make	Qty
16.1	Spindle Air Jet	MFH 3-1/8	Festo	01
16.2	Spindle Air Purge	MFH 3-1/8	Festo	
16.3	ATC Pocket up & Down	JMFH 5- 1/8	Festo	01
16.4	Manual Tool De Clamp	MFH 5-1/8/JMFH 5-1/2	Festo	01
16.5	APC Auto Door	JMFH 5-1/8	Festo	
16.6	CTS Air Jet	MFH 3-1/8	Festo	
16.7	Front Auto Door	JMFH 5-1/8	Festo	
		MFH 3-1/8	Festo	
16.8	Pallet Rest Pad Cleaning	MFH 3-1/8	Festo	
16.9	Fixture		Festo	

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MACHINE HISTORY - MECHANICAL

AMS — In Pursuit of Excellence

17	PRESSURE SWITCH FOR PNEUMATIC SYSTEM			
17.1	Operating Range		2 to 15 Kg /cm2	
17.2	Make		SUCO	
17.3	Quantity		1 + 1	
18	Tool de clamp system:		Mechanical / Pneumatic / Hydraulic / Hydro-Pneumatic	
	Function	Make	Model	Qty
18.1	Manual Tool De Clamp - Pneumatic	Festo	DNU 80-80-PPV DNU 100-80-PPV	1
		Rotex	SCN D -100 / 60 SCN D 80 / 80	1
18.2	Manual Tool De Clamp - Hydraulic	Island Hydraulic / AMS	ф40 X 20 X 65	1
18.3	Manual Tool De Clamp - Hydro pneumatic	Preac	PC33-8045-AM02	1
19	PNEUMATIC CYLINDERS USED			
	Function	Make	Model	Qty
19.1	Front Auto Door	Festo		
19.2	APC Auto Door	Festo		
20	TURRET / INDEXING TABLE (OPTIONAL)			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS			8 / 12 /
21	ROTARY TABLE	Make	Model	Sl.No
		NIKKEN / UCAM		
22	TAIL STOCK	Make	Model	Sl.No
23	PNEUMATIC COUNTER BALANCING			
23.1	Make	SMC / Rotex / Festo		
23.2	Cylinder Diameter :	Ø 80 / 100 / 125		
23.3	Air Reservoir used	Yes / No		
24	HYDRAULIC COUNTER BALANCING			
24.1	Make			
24.2	Model	Ø 50 X 32 X 670		
25	ADDITIONAL INFORMATION / REMARKS			
1				
2				
3				

Checked By

ACE MANUFACTURING SYSTEMS LTD

AMS — In Pursuit of Excellence

CHECK LIST - ELECTRICAL

M/C NUMBER:- MLL-719 | DATE:- 28/09/2011

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of servo motors are as per work order	✓	
2	Axes strokes are as per work order. (P1320 onwards)	✓	
3	All servomotors are taking normal current while running	✓	
4	Check Motor ID & other parameters of axis motor (P2020)	✓	
5	Axis cables are routed properly with suitable clamps & ties	✓	
6	Cables are connected properly both at motor & drive end	✓	
7	Lubrication timers & signals are as required to the machine	✓	
8	Rapid rate & feed rate of axes are as per workorder (P1420/1428)	✓	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Specifications of spindle motor /speed /power is as per work order	✓	
2	Speed locking parameters If any are perfectly set (P3735,3736,)	✓	
3	Power locking parameters If any are perfectly set (P4028,4029)	✓	
4	Rigid tap parameters are set (P4016#4,4044,4052,4085,5241,5261)	✓	
5	Spi. running is checked for direction, noise if any, low & high speed	✓	
6	Spindle motor fan rotation /mpcb rating is checked	✓	
7	Spindle orientation is perfectly set (Proximity signal, parameter)	✓	
8	Manual tool declamp in jog is checked in twin arm models	✓	
9	Switching of Contactors for required speed incase of star/delta motor	✓	
10	Check cable connections both at motor & drive end	✓	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for latest tool change program(O9001) of the machine	✓	
2	Check for macro of twin arm machines	✓	
3	Check for Magazine indexing in both directions	✓	
4	Auto tool change is checked both in jog & auto mode	✓	
5	Check for position sw parameters,maco& reference parametres	✓	
6	Proximity sw cables are properly connected with tags & routed neatly	✓	
7	Tool display is matching with atc position	✓	
8	Power OFF conditions - Message and retrievals are checked	✓	

IV.APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for pallet change program(O9003&100) of the machine.		
2	APC is operating manually through mop & in auto through M80		
3	Proximity sw & limit sw signals are checked for condition during different operation		
4	Working of pallet setup sequence	NA	
5	Interrupt conditions checked & initialization procedure is displayed		
6	Working of fixture interlock if any for pallet change is checked		
7	Cables are routed properly with suitable clamps & ties		
8	Check for position sw parameters,maco& reference parametres		

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ACE MANUFACTURING SYSTEMS LTD

CHECK LIST - ELECTRICAL

V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids / pressure switches are working for specified condition		
2	Messages related to pressure sw settingis appearing when disturbed		
3	Suitable element tags & indicators are put		
4	Cables / plugs are terminated & routed properly		
5	FRL unit signals are functioning properly (check FRL counter)	NA	

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	All solenoids /pressure switches are working for specified conditions		
2	Messages related to pressure sw settings appearing when disturbed		
3	Hydraulic Motor / cooling fan direction is checked		
4	Oil level ,clog, temperature sensors if any, are working	NA	
5	Suitable element tags & indicators are put		
6	Cables / plugs are terminated & routed properly		

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	All coolant motors are running in cw direction	✓	
2	Timers for running all coolant / conveyor motors is checked	✓	
3	Float sw signals are functioning properly	✓	
4	Flow sw,clog sw & air blast is functioning properly in case of cts m/c	NA	
5	Door close condition for coolant 'on' is checked if it is a option	✓	
6	Chip/ screw conveyors are working properly in auto & manual mode	✓	
7	Limit sw / proximity sw signals are functioning properly	✓	
8	MPCB used and settings done is as per the requirement	✓	

VIII.ROTARY/ INDEX TABLES

SL.No	PARAMETERS	STATUS	REMARKS
1	Table is checked for commanded positions & shortest path		
2	Related logic is checked for proper running at safe position		
3	Feedback signals are functioning properly	NA	
4	Related displays are marching with table position		
5	Cable/ hoses are routed properly at safe position		

IX.PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	MCB's are proper rated & connected as per circuit diagram	✓	
2	Rated MPCB's are set & connected as per circuit diagram	✓	
3	Contactors' of required rating & connected as per circuit diagram	✓	
4	Transformers of required rating & connected as per circuit diagram	✓	
5	Relay boards of proper rating & connected as per circuit diagram	✓	
6	Cables of proper rating & properly crimped at connection end	✓	
7	Drive cooling fans are working normally	✓	
8	Panel cooler setting & working is checked	✓	
9	Proper identification stickers / tags are fixed for all elements	✓	

CHECKED BY:- | VERIFIED BY:-

10	Cabinet is fully checked for loose connections if any	✓	
11	PCB , drive, I/O module connectors are firmly fixed	✓	
12	Machine operating panel connections are checked and routed	✓	
13	Moisture free beadings for cabinet are properly fixed	✓	
14	Moisture free beadings for operating panel are properly fixed	✓	

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Illuminations of work lamps & indication lamps are satisfactory	✓	
2	Emergency circuit & monitor is functioning properly	✓	
3	Connections inside pendent are checked	✓	
4	Protective covers for the elements are given if required	✓	
5	All channel covers are fixed with proper fasteners	✓	
6	Protective earth is connected for elements which requires earthing	✓	
7	Safety instructions & other standard stickers are fixed on cabinet	✓	
8	PMC timer values are checked as per data sheet	✓	
9	New design changes are incorporated in the machine		

INTERNAL CHECKLIST POINTS

XI. PMC DATA

Instruction: Please note all D-bits which is enabled at the time of shipping .
Also give the details if it is used for special function

D0	7	6	5	(4)	3	2	(1)	(0)	D13	7	6	5	4	3	2	1	0
D1	7	6	5	4	3	2	1	0	D14	7	6	5	4	3	2	1	0
D2	7	6	5	4	3	2	1	0	D15	7	6	5	4	3	2	1	0
D3	7	6	5	4	3	2	1	0	D16	7	6	5	4	3	2	1	0
D4	7	6	(5)	4	3	2	1	0	D17	7	6	5	4	3	2	1	0
D5	7	6	5	4	3	2	1	0	D18	7	6	5	4	3	2	1	0
D6	(7)	6	5	4	3	2	1	0	D19	7	6	5	4	3	2	1	0
D7	7	6	5	4	(3)	2	1	0	D20	7	6	5	4	3	2	1	0
D8	7	6	(5)	4	3	(2)	1	(0)	D21	7	6	5	4	3	2	1	0
D9	7	6	5	4	3	2	1	(0)	D22	7	6	5	4	3	2	1	0
D10	7	6	5	4	3	2	1	0	D23	7	6	5	4	3	2	1	0
D11	7	6	5	4	3	2	1	0	D24	7	6	5	4	3	2	1	0
D12	7	6	5	4	3	2	1	0	D25	7	6	5	4	3	2	1	0
D																	
D																	

CHECKED BY:- Hari M

VERIFIED BY:- Paul

110	00000000	1	1	✓ 20 Tool
100	00000000	0	10	✓
400	00000010	0	21**	** NO. OF ATC-TOOLS +1 (NOT APLICABLE FOR DISC TYPE ATC
500	00000010	0	21**	✓
600	00000010	0	21**	✓
700	00000010	0	21**	✓
800	00000010	0	21**	✓
000	00000000	3	30	✓

XII.CNC PARAMETERS

Instruction: Please Set the following parameters at the time of shipping .
Also give the details if any special parameter has set

1301#7 = 1 Over travel before move	✓
3103#2 = 1 System alarm history display.	✓
3106#4 = 1 Operation history enable.	✓
3108#6 = 1 Spindle load meter	✓
3111#7 = 0 (NPA) - Message display	✓
3112#0 = 0 Graphics display enable	✓
3112#2 = 1 Operator message history display	✓
3112#3 = 1 Alarm description display in alarm history screen for external alarms	✓
3112#5 = 0 Operator history enable	✓
3117#1 = 1. Spindle position feedback display in dgn 442	✓
3205#4 = 1 In the offset screen offset value CLEAR by soft key to be disabled.	✓
3402#6 =1 (CLR)Set the parameter	✓
1601#4 = 1 (RTO)Set the parameter	✓
1722 = 10/10/10 Set for RTO percentage	✓
1826 =(500,500,500) , 1827=(200,200,200)-check inposition width	✓
6910 - 6965 -check position switches values for ATC and APC	✓
8132.0=1, - TOOL LIFE Parameters	✓
6800#0 , 6801#3 , 6801#7 ,6804#2,6805#4 =1 6810=100, 6811=6, 6844=10, 6845=0	✓
1620 & 1621 Rapid time constant 96, 112, 120 / 88, 96, 96	

Check the following with system:		"1"	"0"	
No.of Ladder steps**	Dgn:1141 # 6	8000	5000 ✓	**N/A for 0 iMD
4th axis option**	Dgn:1279 # 4	Yes	No ✓	**N/A for 0 iMD
Macro executer option	Dgn:1236 # 1	Yes ✓	No	

XIII.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

SL.No	PARAMETERS	REMARKS
1	Take the servo guide back up	✓
2	Check the various communication like RS232C,PCMCIA	✓
3	Edit title,& set time, date other original factory set values	✓
4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	✓
6	Take the back up of following :	✓
	a)CNC parameters b) Pitch error compensation c)PMC parameters	✓
	d) ID- Info f)M- info e) Ladder g) macro i)S-RAM data	✓
7	Make I/O channel '0' for RS232C communication	✓
8	Keep spares inside the cabinet if any (eg.;Fuse)	✓
9	Check relevant circuit diagram of latest edition	✓
10	Ensure all the elements are in good condition before shipping	✓
11	Compare CNC parametrers with standard data & store in backup file	

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

1683

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

M/C NO.: M11-719	DATE: 12/09/11
MODEL: mcv-400	CUSTOMER: matharu Engg works

M/C BACKUP (PATH) : \\arya2\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (20010-11)\FANUC

CNC SYSTEM

0i-MD ☐ 0i-MATE-MD ☑ ☐

PART No A02B-0321-B500

Sl.No.: E109528 61 ✓

SYSTEM OPTIONS:

β SERVO DRIVE SVPM

7.5i (20/20/40) ☐ 15i (40/40/80) ☑

PART No A06B-6164-H343

Sl.No.: V1166 7409

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ ☐

PART No A06B-____-____

Sl.No.:

α SERVO DRIVE SPM

11i ☐ 15i ☐ 22i ☐ ☐

PART No A06B-____-____

Sl.No.:

α SERVO DRIVE SVM2

40/40i ☐ 80/80i ☐ 160/160i ☐

PART No A06B-____-____

Sl.No.:

α SERVO DRIVE SVM1(1'st)

40i ☐ 80i ☐ 160i ☐

PART No A06B-____-____

Sl.No.:

α SERVO DRIVE SVM1(2'nd)

40i ☐ 80i ☐ 160i ☐

PART No A06B-____-____

Sl.No.:

FANUC I/O MODULE / I/O CARD

I/O MODULE WITH MPG ☑	I/O CARD WITH MPG ☐	I/O CARD WITHOUT MPG ☐
PARTNo.: A02B-0309-C001	PART No.: A20B-2002-0520	PART No.: A20B-2002-0521
Sl No. E115D5747	Sl No.	Sl No.

AC REACTOR1

0.14mH/62A ☑ 0.07mH/110A ☐ ☐

PART No A81L-6164-H348

Sl.No.: RT1140974

CHECKED BY:- [signature]

VERIFIED BY:- [signature]

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

SERVO MOTOR - X AXIS

β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0038-B103 #100

Sl.No.: C11B4B388

SERVO MOTOR - Y AXIS

β8/3K ☐ β12/3K ☑ β22/3K ☐ ☐

PART No.: A06B-0038-B103

Sl.No.: C11B7246

SERVO MOTOR - Z AXIS

β12/3K ☐ β22/3K ☑ β22/2K ☐ ☐

PART No.: A06B-0082-B403

Sl.No.: C11A5DB1

B AXIS DETAILS

ROTARY TABLE ☐ INDEX TABLE ☐ ☐

DRIVE: βSiV-20 ☐ βSiV-40 ☐

PART No.: A06B-____-H002

Sl.No.:

MOTOR: β4/4K ☐ β8/3K ☐

PART No.: A06B-0____-____

Sl.No.:

REACTOR: A81L-0001-0018#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

PMM AXIS DETAILS

SERVO ARM ☐ SERVO MAG ☐ TILTING TABLE ☐

DRIVE: βSiV-20 ☐ βSiV-40 ☐

PART No.: A06B-____-H002

Sl.No.:

MOTOR: β4/4K ☐ β8/3K ☐

PART No.: A06B-0____-____

Sl.No.:

REACTOR: A81L-0001-0018#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

SPECIAL INFORMATIONS:

CHECKED BY:- VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

ISOLATTION SWITCH (SW1) — AC 3 Ph 415V ☑

40A	63A	100A
☑	☐	☐

MAKE: SALZER ☑ ABB ☐ ☐

TYPE:

ISOLATTION TRANSFORMER (TFR1) — PRIMARY- 415V/3Ph — SECONDARY - 220V/3Ph

10KVA	12KVA	15.5KVA	
☐	☑	☐	☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 2349-08-11

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A	32A	40A	
☑	☐	☐	☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

SECONDARY MCB (F19) — AC 3 Ph 220V / 2POLE ☑

40A	50A	63A	
☐	☑	☐	☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

DRIVE CONTROL MCB (F20) — NA — AC 3 Ph 220V / 3POLE ☐

1A	3A		
☐	☐	☐	☐

MAKE: SCHNEIDER ☐ MERLIN GERIN ☐

TYPE:

SPINDLE MOTOR FAN MPCB (F56) — AC 3 Ph 220V / 3POLE ☑

0.1-0.16A	0.25-0.4A	
☑	☐	☐

MAKE: ABB ☐ SPRECHER &SCHU ☑

MPCB SETTING: 0.13A

PSM CONTACTOR (K19) — 2 30V AC ☑

CA3-30-10	A30-30-10	CA3-37-N-11	
☐	☐	☑	☐

MAKE: ABB ☐ SPRECHER &SCHU ☑

TYPE:

REGULATED POWER SUPPLY DETAILS

PS1 ☑ PS2 ☑ PS3 ☐ — OUTPUT: 24V DC 10A ☑ 24V DC 20A ☐

MAKE: TELEMECANIQUE(ABL 1RPM24100) ☐ OMRON(S8VE 24024) ☑ KB ELECTORNICS ☐

Sl. No.: PS1: PS2: PS3:

ISOLATTION TRANSFORMER (TFR4) — PRIMARY- 415V/1Ph — SECONDARY - 200V/1Ph

900VA	1.5KVA	
☑	☐	☐

MAKE: SALZER ☐ T.S INTERNATIONAL ☑

SL.No: 1240-08-11

PRIMARY MCB (F16) — AC 3 Ph 415V / 2POLE ☑

1A	3A	
☐	☑	☐

MAKE: SCHNEIDER ☑ MERLIN GERIN ☐

TYPE:

SPECIAL INFORMATIONS: ① F58 → 6 amps ② F59 → 2 amps } For TFR4 Secondary

CHECKED BY:- RSM

VERIFIED BY:- RSM

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

AMS — In Pursuit of Excellence

CONTROL CIRCUIT

SEALED KEY BOARD/MCP: Sl.No.: 1683

TYPE: (YSL-AMS-SKB-2-REV1-10/09) ☑ YSL-SPARK-MCP2-REV2-05/05 ☐

ATC BRAKE RELAY (PCB 1) :

TYPE: 1 SSR RELAY -YSL-SSRP-BR-3-REV1-06/10 ☐ 2 SSR RELAY - YSL-S26-009-77-REV2-08/10 ☑

I/O DISTRIBUTON PCB (PCB 2) : Sl.No.:

TYPE: YSL-S26-009-56-REV0-04/08 ☑ YSL IFB- 50-FCB -TB ☐

CNC ON/OFF RELAY (PCB 3) :

TYPE: YSL -SSR-CNC ON / OFF-REV8- 12/09 08/10 ☑

Z AXIS BRAKE RELAY (PCB 5) :

TYPE: YSL-SSRP-BR3-REV1-06/10 09/09 ☑

Z AXIS BRAKE CURRENT SENSINGRELAY (CSR1) :

TYPE: YSL ELECTRONICS (CSR1-2) ☑

AC RELAYS: Z BRAKE/EMG (ACR1) ☑ ATC/CAB. THERMAL(ACR2) ☑ I. TABLE THERMAL(ACR3) ☐

MAKE: OMRON ☑ ELESTA ☐

RELAY BOARD(STD-X20) 1RELAY(ELECTRO MAGNETIC-DC) ☑

TYPE: YSL-UMK-1RM-24V- DC-CN-FI-2-REV1-12/09 ☑

RELAY BOARD(STD-X31) 1RELAY(ELECTRO MAGNETIC-DC/AC) ☑

TYPE: YSL-S26-009-082-REV0-08/09 ☑

RELAY BOARD(APC-X25) 8 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-064-REV3-08/08 ☐

RELAY BOARD(FIXTURE-X26) 2 / 4 / 8 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-062-REV3-08/08 ☐ S26-009-063-REV3-08/08 ☐ S26-009-064-REV3-08/08 ☐

RELAY BOARD(TOOL DECLAMP - X33) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

RELAY BOARD(- X) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV3-08/08 ☐

EMERGENCY CONTACTOR (K22)

CS8C-40E ☑	KC6-40E ☐	MAKE: ABB ☐	24V DC ☑ SPRECHER &SCHU ☑
CHECKED BY:-		TYPE: VERIFIED BY:-	

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POWER FAILURE DETECTION MODULE (PFDM) :		24V DC ☐
	MAKE:	YSL-3PFMR-2 ☐
CAPACITOR BANK (DP1) :		24V DC ☑
	MAKE:	YSL-PF-CHU-1 ☑

SINGLE PHASE POWER SUPPLY CIRCUIT

MCB (F18)		AC 1 Ph 230V / 1POLE ☑
10A ☑ ☐ ☐	MAKE: SCHNEIDER ☑	MERLIN GERIN ☐
	TYPE:	
PANEL COOLER:	POWER: 500W ☐	1000W ☑
MAKE: ADVANCE (TURBO1000R33) ☑	ADVANCE (TURBO500R50) ☐	ACE (IGLOO - 750) ☐
Sl.No.: 38886-08-11		

UTILITIES (CABINET):

CABINET LAMP:	PEARL(CFL-9W) ☑	PHILIPS(CFL-9W) ☐ ☐
GANG BOX:	2 SWITCHES + 1SOCKET + I INDICATOR ☐	
DOOR INTERLOCK:	LMIT SW (~~NG1RS-510~~ ZLDA01D) 1 / ~~2~~ Nos. ☑	
	F66 → ~~OVERRIDE KEY SW~~ ☑ 3 nos	

UTILITIES (SWITCHGEARS):

ADD ON BLOCK(PCB5/PSM(K19) INTERLOCK):	SPC&SCH(CS7-PV) ☐	ABB ☐
ADD ON BLOCK(STAR DELTA INTERLOCK):	SPC&SCH(CS7-PV) ☐	ABB ☐
ADD ON BLOCK(HYDRAULIC FAN POWER):	SPC&SCH(CS8-P22Z) ☐	ABB(CAF6-20-M) ☐
ADD ON BLOCK(AUTO DRAIN SOL):	SPC&SCH(CS8-P22Z) ☑	ABB(CAF6-20-M) ☐

SPINDLE STAR / DELTA CIRCUIT

CONTACTORS (K20 / K21) N/A		230V DC ☐
CA3-72N-L11 ☐ CA7 - 85 ☐ ☐	MAKE: ABB ☐	SPRECHER &SCHU ☐
	TYPE:	
RELAY BOARD(X32)		2 RELAYS (SSR TYPE) ☐
TYPE: S26-009-80-CN--F-24DC-230AC ☐		

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MAGAZINE MOTOR POWER

MPCB (F1)	1-1.6A ☑	☐	CONTACTOR (K1/K2)	CA8-9C ☑	VBC6-30-01 ☐		
MPCB SETTING:	1A ☑	☐	MOTOR POWER:	1.1KW/1.5A ☑	1.9KVA/2.3A ☐		
SOLID STATE RELAY (SSR1):			MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☑				

ARM MOTOR POWER

MPCB (F2)	1-1.6A ☑	☐	CONTACTOR (K3)	CA8-9C ☑	BC6-30-01 ☐		
MPCB SETTING:	1A ☑	☐	MOTOR POWER:	1.1KW/1.5A ☑	2.3KVA/3.2A ☐		
SOLID STATE RELAY (SSR3):			MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST) ☑				

LUBRICATION MOTOR POWER

MCB (F3)	1A/3P ☑	MPCB(0.63-1A) ☐	CONTACTOR (K4)	CA8-9C ☑	BC6-30-01 ☐		
			MOTOR POWER:	0.12KW/0.6A ☐	☐		

FLOOD COOLANT MOTOR POWER

MPCB (F4)	0.63-1A ☑	2.5-4A ☐	CONTACTOR (K5)	CA8-9C ☑	BC6-30-01 ☐		
MPCB SETTING:	0.9A ☑	3.4A ☐	MOTOR POWER:	0.37KW/0.9A ☑	1.5KW/3.4A ☐		
SURGE SUPRESSOR(RC4):			YSL-NIR-RC3-1/022-575VAC ☑				

CHIP PUSH COOLANT MOTOR POWER

MPCB (F5)	0.63-1A ☐	2.5-4A ☑	CONTACTOR (K6)	CA8-9C ☑	BC6-30-01 ☐		
MPCB SETTING:	0.9A ☐	3.4A ☑	MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☑		
SURGE SUPRESSOR(RC5): YSL-NIR-RC3-1/022-575VAC ☑			COMMON CONTACOR KIT - PUSH / FLUSH(VBC6) ☑				

CHIP FLUSH COOLANT MOTOR POWER

MPCB (F6)	0.63-1A ☑	2.5-4A ☐	CONTACTOR (K7)	CA8-9C ☑	BC6-30-01 ☐		
MPCB SETTING:	0.9A ☑	3.4A ☐	MOTOR POWER:	0.37KW/0.9A ☑	1.5KW/3.4A ☐		
SURGE SUPRESSOR(RC6): YSL-NIR-RC3-1/022-575VAC ☑			COMMON MPCB WITH CHIP PUSH ☑				

COOLANT GUN MOTOR POWER

MPCB (F27)	0.63-1A ☑	2.5-4A ☐	CONTACTOR (K25)	CA8-9C ☑	BC6-30-01 ☐		
MPCB SETTING:	0.9A ☑	3.4A ☐	MOTOR POWER:	0.37KW/0.9A ☑	1.5KW/3.4A ☐		
SURGE SUPRESSOR(RC27):			YSL-NIR-RC3-1/022-575VAC ☑				

HIGH PRESSURE COOLANT MOTOR POWER

MPCB (F9)	4-6.3A ☐	2.5-4A ☐	CONTACTOR (K10)	CA8-12C ☐	BC7-30-01 ☐		
MPCB SETTING:	4.9A ☐	☐	MOTOR POWER:	2.2KW/4.9A ☐	1.5KW/3.4A ☐		
SURGE SUPRESSOR(RC9):			YSL-NIR-RC3-1/022-575VAC ☐				

HYDRAULIC 1 MOTOR POWER

MPCB (F8)	4-6.3A ☐	2.5-4A ☐	CONTACTOR (K9)	CA8-9C ☐	BC7-30-01 ☐		
MPCB SETTING:	2.7A ☐	☐	MOTOR POWER:	1.1KW/2.7A ☐	☐		
SURGE SUPRESSOR(RC8):			YSL-NIR-RC3-1/022-575VAC ☐				

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INDEX TABLE MOTOR POWER

MPCB (F12)	2.5-4A ☐	☐	CONTACTOR(K15/K16)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	3A ☐	☐	MOTOR POWER:	☐ 2.3KVA/3.2A	☐
SOLID STATE RELAY (SSR5):			MAKE: ERI ELECTRONICS- SJDA - 482524(DSPST)		☐

CHIP CONVEYOR 1 MOTOR POWER

MPCB (F10)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K11/K12)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	☐	MOTORPOWER:	0.18KW/0.68A ☐	☐
SURGE SUPRESSOR(RC10):			YSL-NIR-RC3-1/022-575VAC		☐

CHIP CONVEYOR 2 MOTOR POWER

MPCB (F11)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K13/K14)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	☐	MOTORPOWER:	0.18KW/0.68A ☐	☐
SURGE SUPRESSOR(RC11):			YSL-NIR-RC3-1/022-575VAC		☐

CONVEYOR DRUM WASH MOTOR POWER

MPCB (F7)	0.63-1A ☐	2.5-4A ☐	CONTACTOR(K8)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	0.9A ☐	3.4A ☐	MOTOR POWER:	0.37KW/0.9A ☐	1.5KW/3.4A ☐
SURGE SUPRESSOR(RC7):			YSL-NIR-RC3-1/022-575VAC		☐

SCREW CONVEYOR 1 MOTOR POWER

MPCB (F13)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K17/K18)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	1.4A ☐	MOTORPOWER:	0.18KW/0.68A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC13):			YSL-NIR-RC3-1/022-575VAC		☐

SCREW CONVEYOR 2 MOTOR POWER

MPCB (F30)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K28/K29)	CA8-9C ☐	VBC6-30-01 ☐
MPCB SETTING:	0.74A ☐	1.4A ☐	MOTORPOWER:	0.18KW/0.68A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC30):			YSL-NIR-RC3-1/022-575VAC		☐

MOTOR POWER (OPTIONAL-1)

MPCB (F)	☐	☐	CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC		☐

MOTOR POWER (OPTIONAL - 2)

MPCB (F)	☐	☐	CONTACTOR (K)	CA8-9C ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC		☐

SPECIAL INFORMATIONS: MPCB'S- SPRECHER &SCHU(KTA7-25S) ☑ ABB (MS 116) ☐

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ATC & SPINDLE DETAILS

SPINDLE MOTOR: MODEL:	β3 ☐	β6 ☑	β12 ☐		☐
MOTOR RPM:	6000 ☐	8000 ☐	10000 ☑		☐
MOTOR POWER:	7.5/5.5 KW				
PART No:	A06B-1445-B103				
Sl. No:	CM6J051C				
GEAR RATIO:	1:1 ☐	1:1.25 ☐	1:1.5 ☑	DIRECT DRIVE ☐	
POWER LIMIT:	IN KW: 7.5/5.5				
SPEED LIMIT:	IN RPM: 60-6000				

PROXIMITY SENSORS-SPINDLE

MAKE; TEKNIC ☐ TYPE: (24V DC - PNP) ☐

SPINDLE ORIENTATION (PXS1) ☑	M12X40 ☐	M12X52 ☐	M12X68 ☐	M12X73 ☑	
TYPE: CONNECTOR ☑ NON CONN. ☐					
TOOOL CLAMP(PXS3) ☑	M12X40 ☐	M12X52 ☐	M12X68 ☐	M12X73 ☑	
TYPE: CONNECTOR ☑ NON CONN. ☐					
TOOOL DECLAMP(PXS4) ☑	M12X40 ☐	M12X52 ☐	M12X68 ☐	M12X73 ☑	
TYPE: CONNECTOR ☑ NON CONN. ☐					
NO TOOL IN SPINDLE(PXS32) ☑	M08X50 ☐	M08X68 ☐	M12X73 ☐	☐	
TYPE: CONNECTOR ☐ NON CONN. ☐					
ONE ROTATION(PXS39) ☑	M12X40 ☐	M12X52 ☐	M12X68 ☐	M12X73 ☑	
TYPE: CONNECTOR ☑ NON CONN. ☐					

PROXIMITY SENSORS-ATC(DISC TYPE)

MAKE; TEKNIC ☐ TYPE: (24V DC - PNP) ☐

MAGAZINE IN POSITION (PXS2) ☐	M12X40 ☑	M12X52 ☐	M12X68 ☐	M12X73 ☐	
TYPE: CONNECTOR ☐ NON CONN. ☑					
MAGAZINE REFERENCE (PXS46) ☐	M12X73 ☐	FG2X05APO24-5000LFLAT ☐		☐	
TYPE: CONNECTOR ☐ NON CONN. ☐					

PROXIMITY SENSORS-ATC(TWIN ARM TYPE)

MAKE; TEKNIC ☑ TYPE:EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☑

MAGAZINE IN POSITION PROXIMITY (PXS2) ☑	MAG REFERENCE PROXIMITY (PXS46) ☑
POCKET UP PROXIMITY (PXS5) ☑	POCKET DOWN PROXIMITY (PXS6) ☑
ARM AT HOME PROXIMITY (PXS7) ☑	☐
TOOL CLAMP START PROXIMITY (PXS58) ☑	TOOL DECLAMP START PROXIMITY (PXS59) ☑

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PNUEMATIC ELEMENTS (SPINDLE + ATC)

MAKE; SUCO ☑	TYPE:(24V DC - NO CONTACT) ☑
MAIN AIR PRESSURE SWITCH (PRS 2) ☑	SPINDLE AIR PRESSURE SWITCH (PRS 101) ☑
AIR PURGE PRESSURE SWITCH (PRS 102) ☐	☐
MAKE; FESTO ☐	TYPE:(24V DC -COIL) ☐
SPINDLE AIR JET SOLENOID (Y1) ☑	SPI AIR PURGE SOLENOID (Y115) ☐
POCKET UP SOLENOID (Y2) ☑	POCKET DOWN SOLENOID (Y3) ☑
TOOL CLAMP SOLENOID (Y51) ☑	TOOL DECLAMP SOLENOID (Y4) ☑
AIR THROUGH NOZZLE SOLENOID (Y99) ☐	☐

HYDRAULIC ELEMENTS (SPINDLE + ATC)

MAKE; YUKEN ☐ REXROTH ☐	TYPE:(24V DC -COIL) ☐
TOOL CLAMP SOLENOID (Y51) ☑	TOOL DECLAMP SOLENOID (Y4) ☑

UTILITIES(SPINDLE + ATC)

TOOL CHANGE POSITION LIMIT SWITCH - (LS3) :	TEKNIC SG-01-R-610 ☑	☐
TOOL DECLAMP PUSH BUTTON :	ILLUMINATED GREEN ☑	☐
TOOL DECLAMP FOOT SWITCH :	☐	☐
MAG TEST MODE KEY SWITCH (KSW7) :	2 POSITION ☐	☐
Coolant Gun sel sw	2POSITION ☑	☐

SPECIAL INFORMATIONS:

AUTO DOOR & CYCLE START DETAILS

MAKE; FESTO ☐	TYPE:(24V DC -COIL) ☐
AUTO DOOR OPEN SOLENOID (Y6) ☐	AUTO DOOR CLOSE SOLENOID (Y7) ☐
AUTO DOOR SAFETYSOLENOID (Y8) ☐	☐
DOOR SWITCHES MAKE; TECNIK ☐ FESTO ☐	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT) ☐ READ SWITCHES (RSW30& RSW31) ☐
CYCLE START SWITCH MAKE; ☐ TELEMECHANIC ☐	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT) ☐ . ☐
DOUBLE HAND CYCLE START BOX MAKE; TECNIK ☐ ☐	CYCLE START PB (MUSHROOM HEAD GREEN) (02 Nos) ☐ EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) ☐
AUTO DOOR OPERATION MAKE; TECNIK ☐ ☐	AUTO DOOR OPEN/CLOSE PB (GREEN) ☐ AUTO DOOR INTERLOCK KEY SW (2 POSITION) ☐

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COOLANT TANK / CONVEYORS DETAIL

UTILITIES (STD COOLANT)

COOLANT LEVEL FLOAT SWITCH (PXS 8) :	FILPRO(LIM 18PF SC)	☐		☐
COOLANT GUN SEL SW :	2 POSITION	☑		☐
FLUSH COOLANT VALVE (Y128) :	AVCON	☑		☐
COOLANT PLUG (HC1) :	16 PIN	☑	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑

UTILITIES (HP COOLANT):

FLOW SWITCH (FLW1) :	IFM (SH0 500)	☐		☐
CLOG SWITCH (FCLG1) :	HIRLEKAR	☐	SALVIN	☐
AIR BLAST SOLENOID (Y5) :	FESTO	☐		☐
M/C DOOR CLOSE LIMIT SW (LS5) :	TEKNIC(NG1 RS 510)	☐		☐

UTILITIES (CHIP CONVEYOR):

CHIP CONVEYOR 1 O/L LIMIT SW (LS12) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR 2 O/L LIMIT SW (LS13) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐
CHIP CONVEYOR PLUG (HC3) :	10 PIN	☐	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☐

UTILITIES (SCREW CONVEYOR):

SCREW CONVEYOR 1 O/L PROXIMITY (PXS20) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR 2 O/L PROXIMITY(PXS104) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐

SPECIAL INFORMATIONS:

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) :	SHRIDHAN	☐		☐
OIL CLOG SWITCH (CLG2) :	HYDAC	☐		☐
HYDRAULIC THERMAL SWITCH (TH4) :	CARE SYSTEM	☐		☐
HYDRAULIC PLUG (HC2) :	16 PIN	☐	24 PIN	☐
MAKE;INDO ELECTRIC ☐ LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☐

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LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH (FL1) :	CENLUB	☑	☐
LUB PRESSURE SWITCH (PRS1) :	SUCO	☑	☐
AUTO DRAIN SOLENOID (Y44) :	ROTEX	☑	☐

MACHINE LAMPS

MAKE:			PATANKAR	☑	AISHWARYA ☐
MACHINE LAMP1 (ML1) :	11W	☐	9W+9W	☑	36W ☐
MACHINE LAMP2 (ML2) :	11W	☐	9W+9W	☑	36W ☐
MACHINE LAMP3 (ML3) :	11W	☐	9W+9W	☐	36W ☐
MAHINE INDICATION LAMP(PL5) :	1 TIER	☐	3 TIER	☑	3 TIER WITH HOOTER ☐
MAKE:			PATANKAR	☑	☐

UTILITIES(MAHINE)

EMERGENCY MONITOR (PCB 12)	YSL-EMG-PCB-4-REV2-07/06 (04/10)		☑
HAND HELD MPG: NA	FANUC	☐	FUTURE(TAIWAN) ☐
	Sl. No.:		
TPM TRACK :	PIONEER (VER 1.077)	☐	☐
	Sl. No.:		
SERVO VOLTAGE STABILIZER:	15KVA	☐	25KVA ☐
INPUT: OUTPUT:	`MAKE: SERVOMAX	☐	☐
	Sl. No.:		
COOLANT / SPINDLE CHILLER UNIT:	MAKE: KAUKAN	☐	☐
POWER:	Sl. No.:		
CONNECTOR : INDO ELECTRIC (10 PIN) ☐	LOCATION: LHS	☐	RHS ☐

FIXTURE DETAIL

MAKE; ☐		PNEUMATIC ☐	HYDRAULIC ☐
STATION 1 CLAMP SOLENOID (Y70) ☐		STATION 1 DECLAMP SOLENOID (Y71) ☐	
STATION 2 CLAMP SOLENOID (Y72) ☐		STATION 2 DECLAMP SOLENOID (Y73) ☐	
STATION 3 CLAMP SOLENOID (Y84) ☐ NA		STATION 3 DECLAMP SOLENOID (Y85) ☐	
STATION 4 CLAMP SOLENOID (Y86) ☐		STATION 4 DECLAMP SOLENOID (Y87) ☐	
MAKE; ☐		PNEUMATIC ☐	HYDRAULIC ☐
STATION 1 CLAMP PRESSURE SW (PRS20) ☐		STATION 2 CLAMP PRESSURE SW (PRS22) ☐	
STATION 3 CLAMP PRESSURE SW (PRS31) ☐		STATION 4 CLAMP PRESSURE SW (PRS32) ☐	

SPECIAL INFORMATIONS: CLAMP PB'S (GREEN) -

DECLAMP PB'S (RED) -

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APC (LINEAR TYPE) DETAIL

MANUAL OPERATION PANEL :	YSL-AMS-APC-PCB-REV0-07/06	☐
PALLET SET UP PB :	LUMINOUS GREEN	☐
EMERGENCY PB :	MUSHROOM HEAD RED	☐

PROXIMITY SENSORS:

MAKE; TEKNIC ☐		TYPE: (24V DC - PNP)			☐
PALLET RECEIVED PROXIMITY(PXS11) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 1 IN PROXIMITY (PXS13) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 1 OUT PROXIMITY (PXS14) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 2 IN PROXIMITY (PXS15) ☐	M12X40	M12X52	M12X68	M12X73	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	
ARM 2 OUT PROXIMITY (PXS16) ☐	M12X40	M12X52	M12X68	M12X72	
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐	

PNEUMATIC SOLENOIDS:

MAKE; FESTO ☐	TYPE: 24V DC -COIL ☐
APC DOOR OPEN SOLENOID (Y12) ☐	APC DOOR CLOSE SOLENOID (Y13) ☐
ALLET PAD CLEAN SOLENOID (Y18) ☐	☐

HYDRAULIC SOLENOIDS:

MAKE; YUKEN ☐ REXROTH ☐	TYPE: 24V DC -COIL ☐
PALLET CLAMP SOLENOID (Y10) ☐	PALLET CLAMP SOLENOID (Y11) ☐
ARM 1 IN SOLENOID (Y14) ☐	ARM 1 OUT SOLENOID (Y15) ☐
ARM 2 IN SOLENOID (Y16) ☐	ARM 2 OUT SOLENOID (Y17) ☐

HYDRAULIC PRESSURE SWITCHES:

MAKE; POLYHYDRON ☐	TYPE: ☐
PALLET CLAMP PRSW (PRS3) ☐	PALLET DECLAMP PRSW (PRS60) ☐

LIMIT SWITCHES:

MAKE; TEKNIC(NG1RSQ 510) ☐	JAI BALAJI (JLSSPR) ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☐	PALLET 2 ON STATION LIMIT SW (LS11) ☐

READ SWITCHES:

MAKE; FESTO ☐	APC DOOR INTERMEDIATE READ SW (RSW 65) ☐
APC DOOR OPEN READ SW (RSW 58) ☐	APC DOOR CLOSE READ SW (RSW 59) ☐

SPECIAL INFORMATIONS:

CHECKED BY:-	VERIFIED BY:-

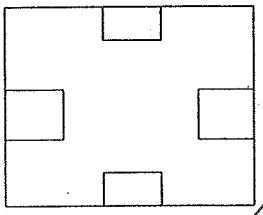
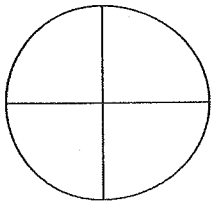
ACE MANUFACTURING SYSTEMS LTD

FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

AMS — In Pursuit of Excellence

MACHINE No. :	CUSTOMER :

I. Motor Mounting Bracket Face Out and CONCENTRICITY for Direct Drive

	FACE OUT (Recommended ≤ 0.015 mm)	CONCENTRICITY (Recommended ≤ 0.010 mm)
	 N/A	

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
1	Spindle Connector Radial Run Out	≤ 0.010 mm	/
2	Motor Current at Maximum RPM	β3 < 15%, β6 / α6 & ABOVE <10%	/
3	CTS Rod Guide Radial Run Out (DDCTS)	≤ 0.010 mm	N/A
4	Spindle Clamping Force	As per DS-DSN-PROD-01	/
5	Spindle Motor Specification	As per Work Order	/
II	**INDIRECT DRIVE**		
1	Motor Pulley Radial Run Out	≤ 0.050 mm	0.03 mm
2	Spindle Pulley Radial Run Out	≤ 0.050 mm	0.03mm
3 a)	Belt Tension	As per DS-DSN-PROD-02	Ok
b)	Belt Specification	Size :	Quantity : 01
4	Motor Current at Maximum RPM	β3 < 20%, β6 / α6 & ABOVE <15%	10%
5	Rotary Joint Radial Run Out (ID CTS)	≤ 0.040 mm	N/A
6	Spindle Clamping Force	As per DS-DSN-PROD-01	800kg.
7	Spindle Motor Specification	As per Work Order	B.6
8	Pulley Ratio	As per Work Order	1:1.25
III	Axis Dead Stopper Value / Butt Referencing Value (As per Drawing)		

	Axis	MOTOR END	BRAKE END
1	X	25.000	21.095
2	Y	21.107	17.392
3	Z	21.261	9.692

IV	% Of axis CURRENT Drawn at 100 % Rapid Rate (Recommended as per DS-QA-02)	
	X - Axis	48%
	Y- Axis	51%
	Z - Axis Up	65%
	Z - Axis Down	29%
V	Accumulator Oil Pressure	As per Circuit Diagram — N/A
	Accumulator Gas Pressure	As per Circuit Diagram — N/A

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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

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S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
VI	ATC ALIGNMENT (TURRET TYPE)		
1	Parallelism of gripper Arm face wrto. X - Axis	≤ 0.050 mm	
2	Tool Ejection Stroke	As per DS - DSN-PROD-01	
3	Swing Lever Gap	As per Drawing	N/A
4	Side Cam Alignment	0.020 mm Maximum	
5	Gap between Draw Bar bush & Roller/Screw	1 ± 0.1 mm	
VII	ATC ALIGNMENT (TWIN ARM)		
1	X - Axis	≤ 0.2 mm	0.02
2	Y - Axis	≤ 0.2 mm	0.02
3	Z - Axis	≤ 0.05 mm	0.01
4	Twin Arm bottom face parallelism in Y-Axis	≤ 0.1 mm	0.1
5	Gap between Draw Bar Bush & Roller	1 ± 0.1 mm	N/A
6	Tool Ejection stroke	As per DS - DSN-PROD-01	1 mm
7	Gap between deflection Pads	0.02 TO 0.050 mm	N/A
VIII	APC ALIGNMENT - LINEAR TYPE		
1	Height difference of rest pads	≤ 0.010 mm	N/A
2	Pallet change Time	As per Work Order	
IX	APC ALIGNMENT- ROTARY TYPE		
1	APC Base parallelism w.r.to. X - Axis	≤ 0.1 mm	
	APC Base parallelism w.r.to. Y - Axis	≤ 0.1 mm	
2	Arm Top parallelism w.r.to. X - Axis	≤ 0.05 mm	
	Arm Top parallelism w.r.to. Y - Axis	≤ 0.05 mm	
3	Arm 1 Alignment ' FACE ' w.r.to X- Axis	≤ 0.02 mm	N/A
	Arm 1 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' FACE ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
4	Offset between Arm 1 & Arm 2 - Face	≤ 0.030 mm	—
	Offset between Arm 1 & Arm 2 - Taper	≤ 0.030 mm	
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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
5	Pallet - 1 Wedge Alignment along X - Axis	≤ 0.020 mm	
	Pallet - 2 Wedge Alignment along X - Axis	≤ 0.020 mm	
6	Gap between Pallet -1 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -1 side & Arm along Z - Axis	0.1 ± 0.02 mm	
	Gap between Pallet -2 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -2 side & Arm along Z - Axis	0.1 ± 0.02 mm	N/A
7	Pallet Sagging - Pallet -1 'W.r.to. X & Y - Axis	≤ 1 mm	
	Pallet Sagging - Pallet -2 'W.r.to. X & Y - Axis	≤ 1 mm	
8	Pallet Height Variation	As per TEST CHART	
9	Backlash between Rack & Pinion at distance of 500 mm	≤ 1 mm	
10	Pallet Change Time	As per Work Order	

S.No.	CHECK POINTS	REMARKS
A	**AXIS**	
1	Ensure geometrical accuracies as per test chart.	OK
2	Ensure plugs are fitted for LM guide ways.	OK
3	Ensure butt reference function.	OK
4	Ensure effective wiping of saddle cover wipers.	N/A
5	Ensure lube timer is set 12 min for HG and 20 min for LM Guide ways.	OK
6	Ensure leveling bolt locknuts are tighten after machine leveling.	OK
7	Check the all axes reference arrows are matched & Direction name plate fixed.	OK
8	Ensure tightness & marking of all screws with Yellow paint marker	OK
9	Ensure proper setting of Limit switches.	OK
B	**ATC**	
1	Check oil level is maximum in ATC gear box.	OK
2	Check tool change position limit switch for proper functioning & band width (Max. 1 mm on each side).	OK
3	Ensure smooth function of Tool change without abnormal/unhealthy noise.	OK
4	Ensure ATC bracket & Magazine Dowelling.	OK

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FINAL ASSEMBLY - INSPECTION REPORT (VMC / CMV) - MECHANICAL

S.No.	CHECK POINTS	REMARKS
C	**SPINDLE**	
1	Check the tightness of push stud.	OK
2	Ensure air jet through all the (4 / 8) holes at oriented spindle position	OK
3	Ensure supply of air to spindle Air Purge system (If provided)	N/A
4	Ensure proper setting of proximities.	OK
5	Ensure smooth running of spindle without abnormal/unhealthy noise	OK
6	Ensure dummy plate fixing on CTS slot in Non CTS machine.	N/A
7	Check the Tool not in spindle proximity message.	N/A
D	**COOLANT**	
1	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe	OK
2	Ensure proper working of float switch (by shifting the sensor upwards).	OK
3	Ensure all the hoses, fittings tightened and routed properly.	OK
4	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).	OK
5	Ensure CTS flow sw.setting by opening the bag filter air vent ball valve.(Optional)	N/A
6	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working (If provided)	N/A
7	Ensure CTS pin locator hole is plugged (If provided)	N/A
8	Ensure that twin arm plunger & CTS pin locator screw do not foul (if provided)	N/A
9	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)	N/A
10	Ensure High pressure air blast after CTS switched off (If provided)	N/A
11	Ensure drum wash in conveyors (If provided)	N/A
12	Check conveyor overload message & Direction (If provided)	N/A
13	Check oil in conveyor gear box. (if provided)	N/A
14	Ensure functioning of screw conveyor in forward & reverse directions (through selector switch).	N/A
15	Ensure splash guard fixed to conveyor.	N/A
E	**APC**	
1	Ensure proper setting of proximities.	
2	Ensure supply of air jet to pallet rest pad cleaning.	
3	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise	N/A
4	Ensure APC base Dowelling.	
5	Check pallet initialization in APC Machine	
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ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
FINAL ASSEMBLY -INSPECTION REPORT (VMC / CMV)- MECHANICAL		
S.No.	**CHECK POINTS**	
F	**HYDRAULIC, PNEUMATIC & LUBRICATION**	
1	Ensure pneumatic main line pressure switch setting (4 bar) & error message are satisfactory	OK
2	Ensure Spindle air jet pressure switch setting (3 bar) & error message are satisfactory	OK
3	Ensure Spindle air purge pressure switch setting (1.2 bar) & error message are satisfactory	N/A
4	Ensure proper setting of oil flow (1 drop/tool change) in FRL unit. & Oil Level low message.	OK
5	Ensure proper routing of FRL drain hose.	OK
6	Ensure identification name plates are fixed on power pack hoses.	N/A
7	Ensure Ball valve fixed to FRL Unit inlet	N/A
8	Ensure pneumatic quick change coupling fixed to 4th axis provision	N/A
9	Ensure no oil leakage in hydraulic valves & fittings	N/A
10	Ensure hydraulic flow control valve grub locking in APC machine.	N/A
11	Ensure pneumatic counter balance bracket Dowelling.	N/A
G	**GUARDING ASSEMBLY**	
1	Ensure all internal guards are assembled.	OK
2	Ensure smooth movement of sliding door without play and unhealthy noise	OK
3	Ensure Nylock nuts are fixed to door rollers.	OK
4	Ensure ATC door LM rail plugs fixed.	N/A
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	N/A
H	**ANY OTHER POINTS**	
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ACE MANUFACTURING SYSTEMS LTD

GEOMETRICAL TEST CHART(VMC / CMV)

PERMISSIBLE TOLERANCE AS PER TEST CHART (DS-DSN-PROD-TC-01)

1. MACHINE LEVEL READING:

Along X-Axis	Along Y-Axis
0	0
0	0
0	0

2. TABLE TOP READINGS:

Along X-Axis		
0	-5	-10
0	-2	-5
0	+2	+5

Along Y-Axis		
0	0	0
-5	-2	-3
+10	-5	-2

3. T SLOT READING:

0	+10

PALLET - II

PALLET - II

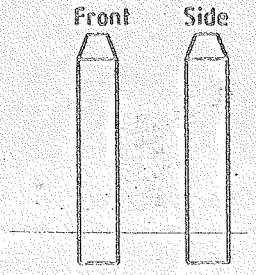
Along X-Axis	
0	-5

Along Y-Axis

4. COLUMN READINGS :

FRONT	SIDE
+2	-5
0	0

5! MANDREL READINGS:

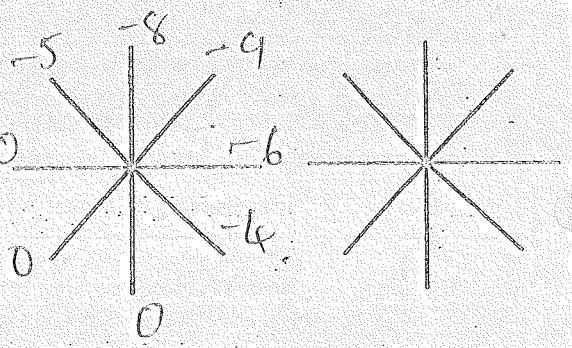


RUNOUT:

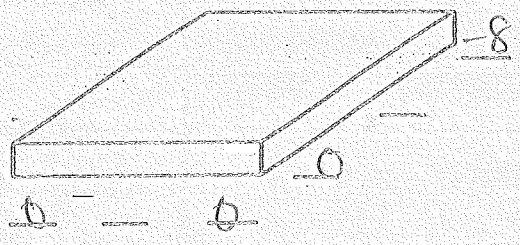
a. TOP :
1.

b. BOTTOM :
1.
2.
3.
4.

6. TURN AROUND READING:



7. SQUARENESS READING (X&Y):



8. PALLET HEIGHT VARIATION:

Pallet-1:

Pallet-2 :

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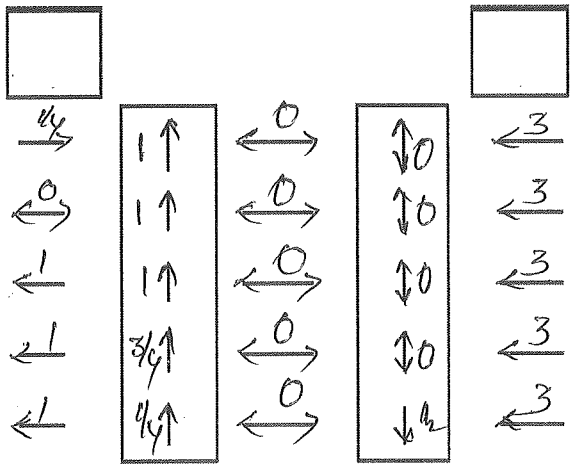
ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS *In Pursuit of Excellence*

MACHINE NUMBER: VMC-709 **DATE:** 05-09-2011

I. MACHINE BASE LEVELING: Least count of the Instrument: 1 div = 0.020mm / meter



SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	1 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1 div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.002		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.005		
6	Ball screw Run out at Motor end	±0.010	0.004		
7	Ball screw Run out at Brake end	±0.010	0.006		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	0.7	0.7	0.7
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
15	Final slide Torque – box guide way	2.5 to 3 N-m	—	—	—
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	—	—	—
17	Variation in torque for final slide	10%	—	—	—
18	Remarks if any :				

ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING, 2 GUIDEWAYS

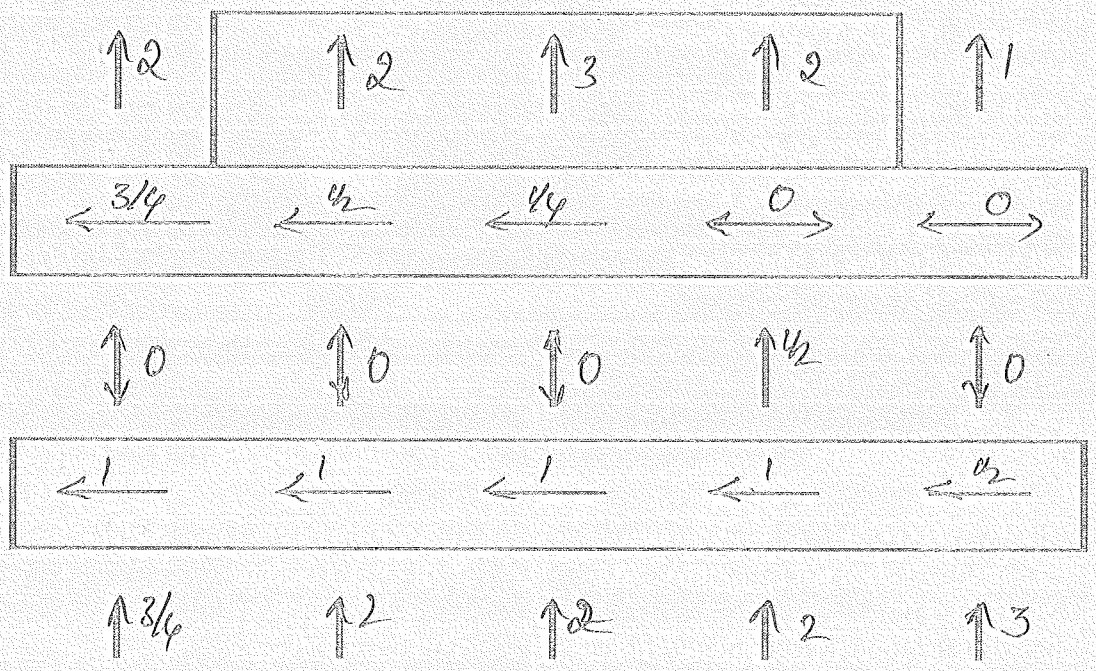
AMS *In Pursuit of Excellence*

MACHINE NUMBER: MH-719 DATE: 05-09-2011

SL NO	Check Points	Remark
1	Ensure all leveling screws & locknuts are fixed to the base.	OK
2	Ensure guide way butting face blue matching is minimum 70%	OK
3	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	OK
4	Ensure position of LM block ground surface butted to slide butting face.	OK
5	Ensure required height blocks are fitted to the base as per Work Order / Planning chart.	OK
6	Ensure plugs are fitted for L M guide way	OK

DATE: 07-09-2011

II. SADDLE LEVEL READING: Least count of the Instrument: 1 div = 0.020mm / meter



Remarks if any :

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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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MACHINE NUMBER: MH-718 **DATE:** 07-09-2011

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	2 div / meter	2 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1 div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.005		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.005		
6	Ball screw Run out at Motor end	±0.010	0.008		
7	Ball screw Run out at Brake end	±0.010	0.006		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.2	1.2	1.2
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
15	Final slide Torque – box guide way	2.5 to 3 N-m	—	—	—
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	—	—	—
17	Variation in torque for final slide	10%			

Remarks if any :

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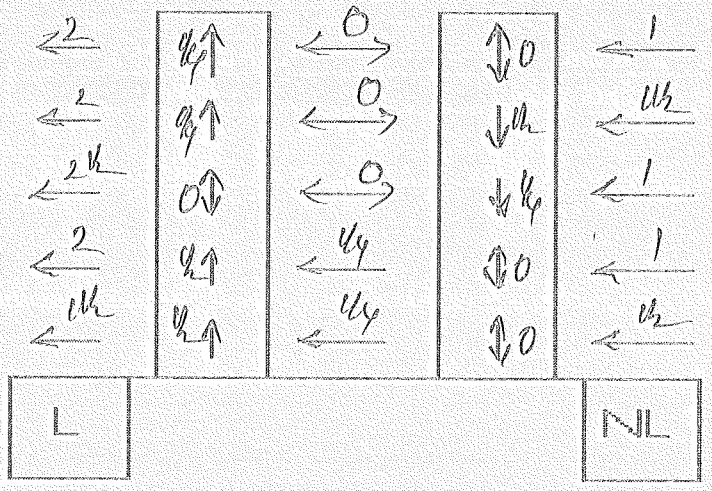
ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: MU-719 DATE: 07-09-2011

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene	OK
2	Ensure guide way butting face & saddle bottom blue matching is minimum 80%	OK
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	—
4	Ensure proper taper matching of jib. ((H G m/c only)	—
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	OK
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	OK
7	Ensure Equal gap between ball nut & nut housing bore.	OK
8	Ensure Nut housing alignment is with in 0.010 mm/meter	OK
9	Ensure White Prussian matching between nut housing & ball nut butting face is80%	OK
10	Ensure bearing housing Dowelling and grub locking.	OK
11	Ensure Ball screw lock nut grub locking with lock tite.	OK
12	Ensure guide way wipers are fixed. (H G m/c only)	—
13	Ensure Nylon stoppers are fixed as per specification.	OK
14	ensure plugs are fitted for lm guide way & grub screws are fixed to saddle mounting holes	OK

DATE: 06-09-2011

III. COLUMN READING: Least count of the Instrument: 1 div = 0.020mm / meter



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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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MACHINE NUMBER: MH-719 **DATE:** 06-09-2014

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	1 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	1 div		
3	Leveling - Individual (Transverse)	3 div / meter	2 div		
4	Straightness of locating guide way	± 0.005	0.002		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.004		
6	Ball screw Run out at Motor end	±0.010	0.008		
7	Ball screw Run out at Brake end	±0.010	0.006		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025		
11	Mandrel radial run out	±0.010	0.006		
12	Mandrel reading at Top (parallelism wrt.head stock movement)	±0.010	0.005		
13	Mandrel reading at side (parallelism wrt.head stock movement)	±0.010	0.008		
			M.E	CEN	B.E
14	Ball screw Torque	0.6 to 1.8 N-m	0.8	0.8	0.8
15	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
16	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
17	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0	—	—	—
18	Final slide Torque – box guide way	2.5 to 3 N-m	—	—	—
19	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.25	1.25	1.25
20	Variation in torque for final slide	10%			

Remarks if any :

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ACE MANUFACTURING SYSTEMS LTD
INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: MU-719 DATE: 06-09-204

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	OK
2	Ensure guide way butting face, head stock bottom face,& column bottom face blue matching is minimum 80%	OK
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	—
4	Ensure proper matching of jib taper. ((H G m/c only)	—
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	OK
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	OK
7	Ensure Equal gap between ball nut & nut housing bore.	OK
8	Ensure nut housing alignment is with in 0.010 mm	OK
9	Ensure White Prussian matching between nut housing & ball nut butting face is 80 %	OK
10	Ensure spindle cartridge butting face blue matching is 80%	OK
11	Ensure head stock is fixed as per requirement (ID, DD, and DD CTS).	OK
12	Ensure bearing housing Dowelling and grub locking.	OK
13	Ensure Ball screw lock nut grub locking with locktite.	OK
14	Ensure guide way wipers are fixed. (H G m/c only)	—
15	Ensure Nylon stoppers are fixed as per specification.	OK
16	Ensure machine lifting tackles are fitted.	OK
17	Ensure plugs are fitted for lm guide way & grub screws are fixed to head stock mounting holes	OK

MACHINE NUMBER: DATE:

IV. SIDE ARM ALINGMENT READING: (Turret Type ATC only)

SL NO	DESCRIPTION	Recommended (mm)	ACTUAL (mm)
1	Shaft reading at Top (longitudinal)	±0.050	
2	Shaft reading at side (transverse)	±0.030	
3	Square ness of locating side	±0.010	N/A
4	Distance between Z axis guide way top surface & side arm bore centre	REFER AS PER DRAWING NO	
5	Distance between sidearm front butting face on column & side arm bore centre	REFER AS PER DRAWING NO	

Remarks if any :

ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS *In Pursuit of Excellence*

MACHINE NUMBER: MM-709 **DATE:** 24-09-2014

RECOMMENDED TOLERANCE AS PER DS – DSN – TC – PROD – 01 – R0

V. LEVELLING OF MACHINE: **L.C of Instrument: 1 div = 0.020mm / m**

Along X-axis			Along Y-axis		
Motor End	Middle	Brake End	Motor End	Middle	Brake End
1/4 →	0 ↔	← 1/2	↑ 1/4	↕ 0	↑ 1/2
1/2 →	0 ↔	← 3/4	↑ 1/2	↕ 0	↑ 1/4
1/2 →	0 ↔	← 1/2	↕ 0	↓ 1/4	↕ 0

VI. TABLE TOP READING ALONG X- AXIS:

0	-2	-4	-5
0	-2	-3	-5
0	-2	-3	-3
0	-1	+1	+3

VII. TABLE TOP READING ALONG Y- AXIS:

0	0	0	0
0	0	0	-1
-1	-1	-1	+1
-3	-2	-1	+4

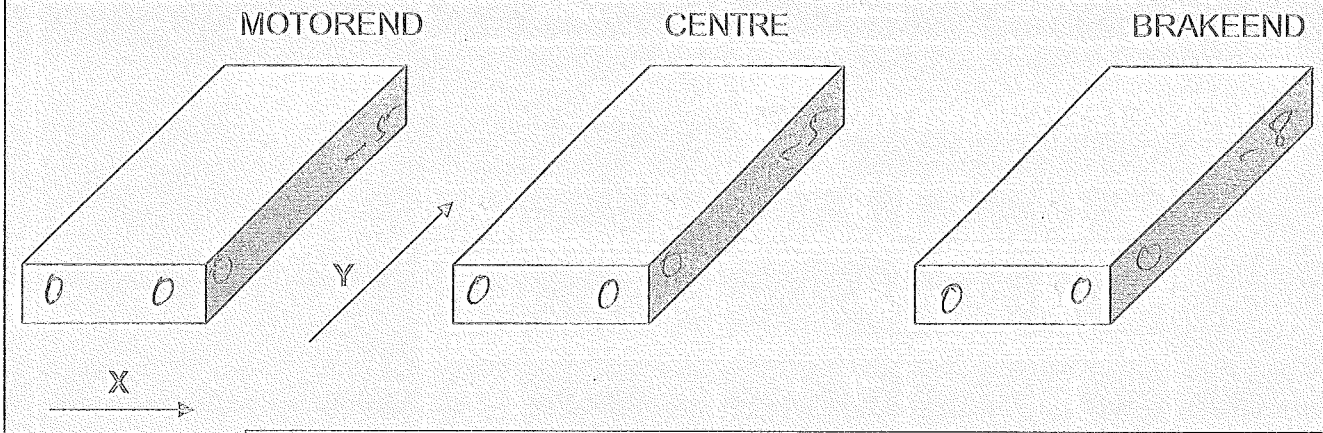
MACHINE NUMBER: 0111-719 DATE: 15-09-2011

VIII. PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT AT FRONT & REAR SURFACE:

0	-4	-6

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air	OK
2	Ensure table bottom face blue matching IS minimum 70%	OK
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	—
4	Ensure proper matching of jib taper. ((H G m/c only)	—
5	Ensure position of LM block ground surface butted to slide butting face.	OK
6	Ensure Equal gap between ball nut & nut housing bore.	OK
7	Ensure nut housing parallelism is with in 0.010 mm	OK
8	Ensure Blue matching between nut housing & ball nut butting face is 70%	OK
9	Ensure Table Tenon slot size as per specification	OK

IX. SQUARENESS OF Y –AXIS MOVEMENT WITH RESPECT TO X – AXIS MOVEMENT:



Remarks if any :

ACE MANUFACTURING SYSTEMS LTD

INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

AMS
In Pursuit of Excellence

MACHINE NUMBER: MM-719 **DATE:** 16-09-2014

X. COLUMN READING (INITIAL):

XZ PLANE(SIDE)

500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

YZ PLANE (FRONT)

500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

COLUMN READING - FINAL:

XZ PLANE(SIDE)

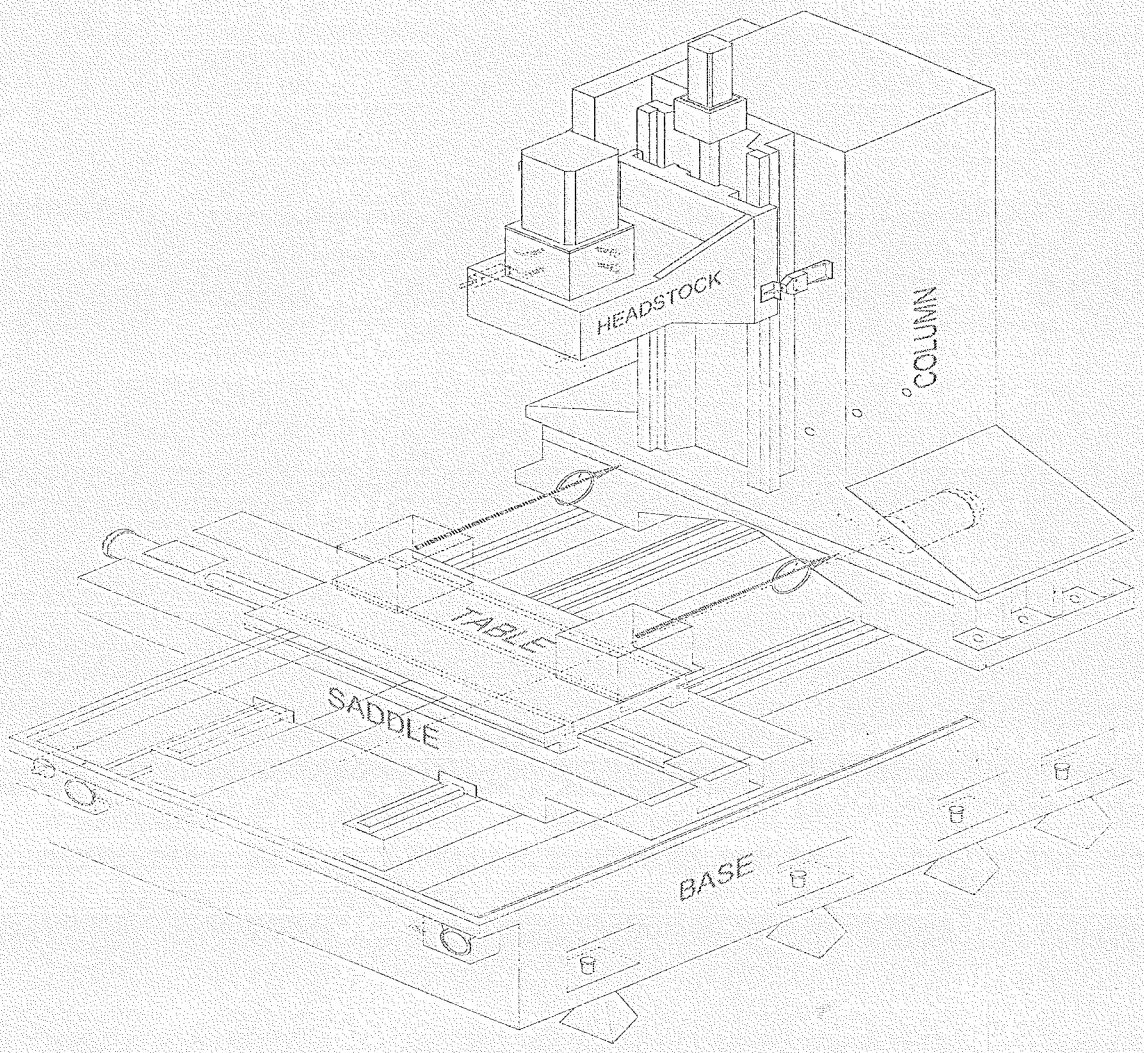
500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

YZ PLANE (FRONT)

500	
450	
400	
350	
300	
250	
200	
150	
100	
50	
0	

MACHINE NUMBER: DATE:

XI. COLUMN ALINGMENT READING WITH RESPECT TO X-AXIS MOVEMENT:
PERMISSIBLE VALUE = 0.30MM/METRE



Remarks if any :

SIGN & DATE				10...10	F-QA-IPR-BBA-03-R0
	PREPARED	CHECKED	APPROVED	PAGE NUMBER	FORMAT NUMBER

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-719-X.RTL

Machine:MCV-400.
Date:15:07 Sep 24 2011
Axis:X.

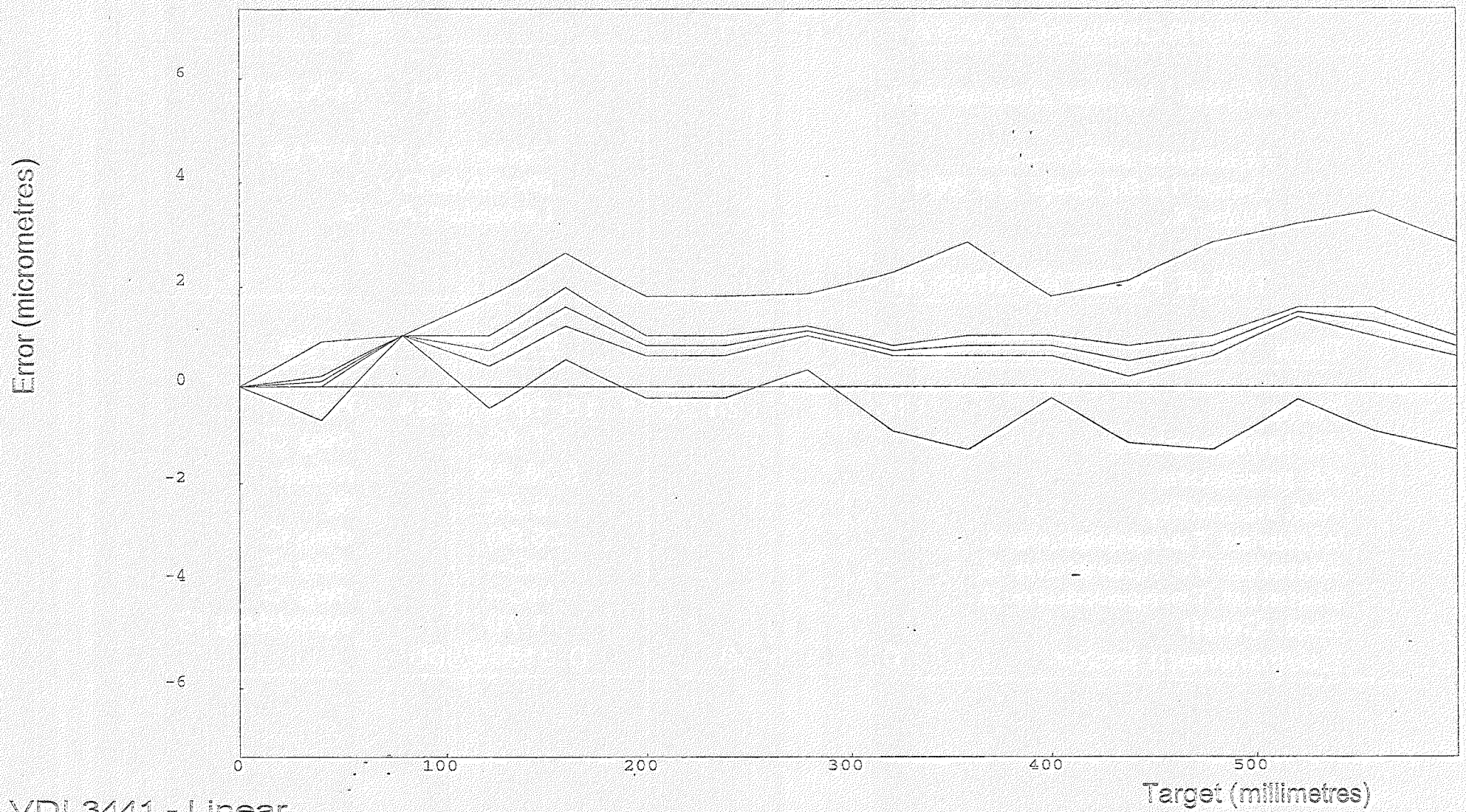
Serial No:M11-719.
By:QAS/DVS.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	15:07 Sep 24 2011	15:16 Sep 24 2011	
Air temperature :	27.45	27.45	deg C
Air pressure :	906.40	906.40	mbar
Rel. humidity :	46.26	46.26	%
Material Temperature 1 :	25.90	25.90	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639866	0.31639866	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:X.	U max : 0.800
Serial No:M11-719.	Location:AMS.	P : 4.764
Date:15:07 Sep 24 2011	U mean : 0.362	Ps max : 3.764
By:QAS/DVS.	Ps mean : 2.163	Pa : 1.600

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-719.
Date:15:07 Sep 24 2011 By:QAS/DVS.
Axis:X. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-719-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (39.9990 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 2 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 3 (+) (79.9980 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 3 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 4 (+) (119.9970 millimetres)
1) 1.0 2) 0.0 3) 1.0 4) 0.0
5) 0.0
Target 4 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 5 (+) (159.9960 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0
Target 5 (-)
1) 2.0 2) 2.0 3) 2.0 4) 2.0
5) 2.0

Target 6 (+) (199.9950 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 6 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 7 (+) (239.9940 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 7 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 8 (+) (279.9930 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 8 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0

Target 9 (+) (319.9920 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 9 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 10 (+) (359.9910 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 10 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0

Target 11 (+) (399.9900 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 11 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 12 (+) (439.9890 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 1.0
Target 12 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (479.9880 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 13 (-)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0

Target 14 (+) (519.9870 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 2.0
5) 2.0
Target 14 (-)
1) 1.0 2) 1.0 3) 2.0 4) 2.0
5) 2.0

Target 15 (+) (559.9860 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 1.0
5) 2.0
Target 15 (-)
1) 1.0 2) 1.0 3) 2.0 4) 2.0
5) 2.0

Target 16 (+) (599.9850 millimetres)
1) 0.0 2) 0.0 3) 1.0 4) 1.0
5) 1.0
Target 16 (-)
1) 0.0 2) 1.0 3) 1.0 4) 2.0
5) 1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-719-Y.RTL

Machine:MCV-400.
Date:15:50 Sep 24 2011
Axis:Y.

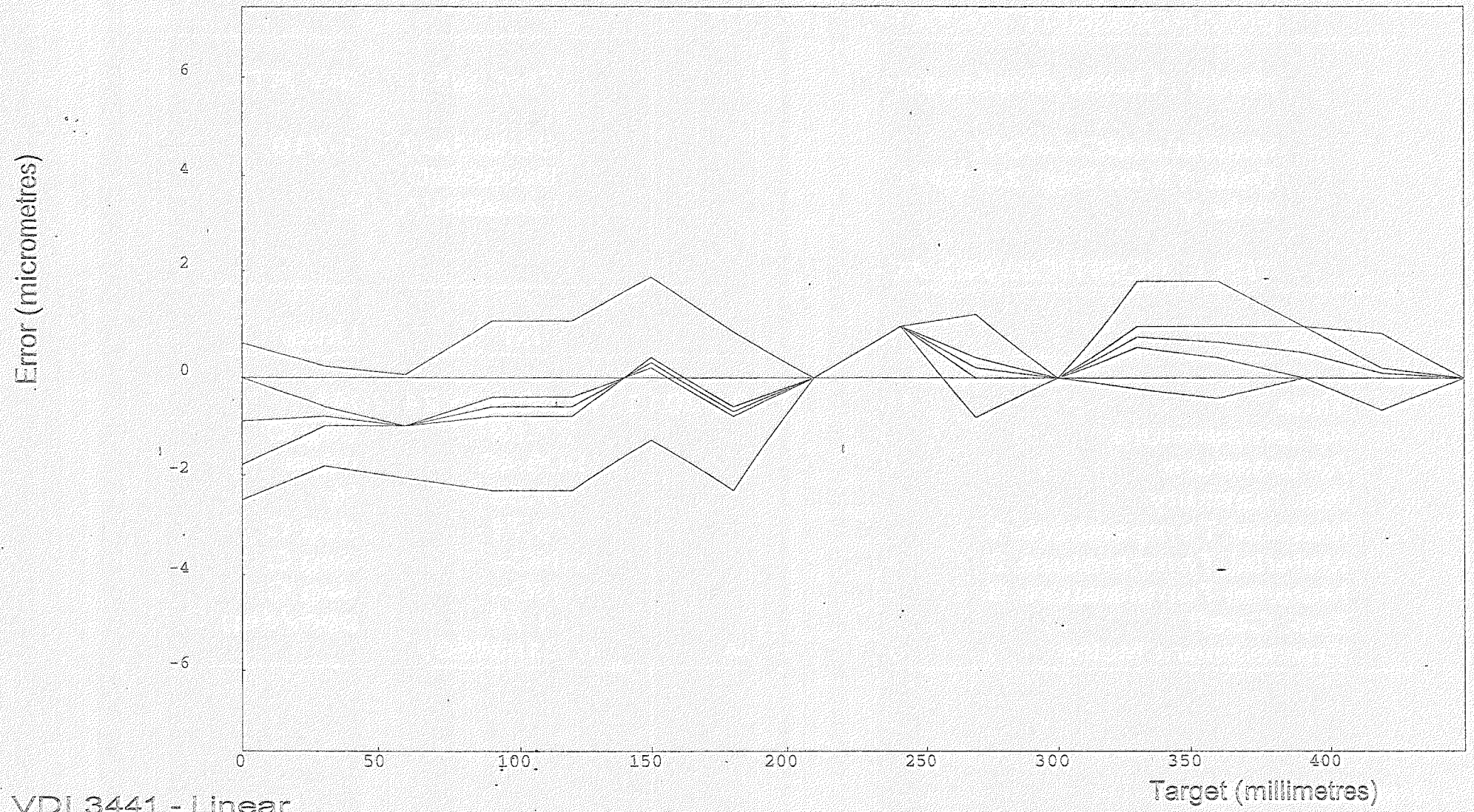
Serial No:M11-719.
By:QAS/DVS.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	15:50 Sep 24 2011	16:00 Sep 24 2011	
Air temperature :	27.58	27.52	deg C
Air pressure :	906.40	906.40	mbar
Rel. humidity :	46.06	47.35	%
Material Temperature 1 :	26.64	26.71	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639594	0.31639567	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Y.	U max : 1.800
Serial No:M11-719.	Location:AMS.	P : 4.363
Date:15:50 Sep 24 2011	U mean : 0.375	Ps max : 2.985
By:QAS/DVS.	Ps mean : 1.457	Pa : 2.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-719.
Date:15:50 Sep 24 2011 By:QAS/DVS.
Axis:Y. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-719-Y.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) -1.0 2) -2.0 3) -2.0 4) -2.0
5) -2.0

Target 2 (+) (29.9990 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 3 (+) (59.9980 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 3 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -2.0

Target 4 (+) (89.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0
Target 4 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (119.9960 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) 0.0
5) -1.0
Target 5 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (149.9950 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 1.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (179.9940 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (209.9930 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 9 (+) (239.9920 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0
Target 9 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 10 (+) (269.9910 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 10 (-)
1) 1.0 2) 0.0 3) 1.0 4) 0.0
5) 0.0

Target 11 (+) (299.9900 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 12 (+) (329.9890 millimetres)
1) 0.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0
Target 12 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 13 (+) (359.9880 millimetres)
1) 0.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (389.9870 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (419.9860 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 0.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0

Target 16 (+) (449.9850 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 16 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-719-Z.RTL

Machine:MCV-400.
Date:16:38 Sep 24 2011
Axis:Z.

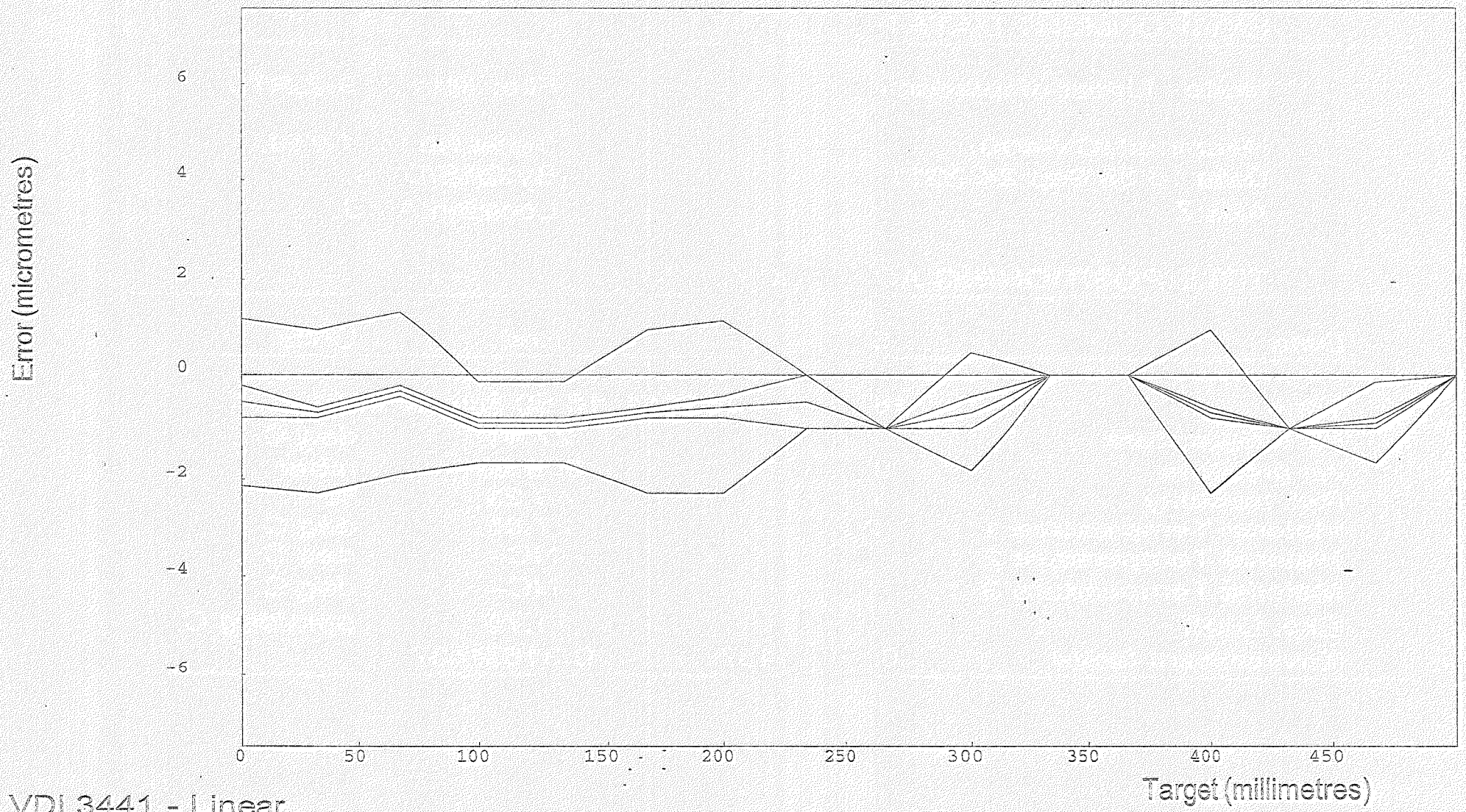
Serial No:M11-719.
By:QAS/DVS.
Location:AMS.

TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	16:40 Sep 24 2011	16:50 Sep 24 2011	
Air temperature :	27.53	27.41	deg C
Air pressure :	906.80	906.80	mbar
Rel. humidity :	47.53	48.23	%
Material Temperature 1 :	27.00	27.05	deg C
Material Temperature 2 :	--NC--	--NC--	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639457	0.31639437	

VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Z.	U max : 1.000
Serial No:M11-719.	Location:AMS.	P : 3.585
Date:16:38 Sep 24 2011	U mean : 0.250	Ps max : 2.985
By:QAS/DVS.	Ps mean : 1.455	Pa : 1.000

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-719.
Date:16:38 Sep 24 2011 By:QAS/DVS.
Axis:Z. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-719-Z.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0
Target 1 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 2 (+) (33.3330 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 3 (+) (66.6660 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0

Target 4 (+) (99.9990 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 4 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 5 (+) (133.3320 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 5 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (166.6650 millimetres)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0
Target 6 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 7 (+) (199.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0
Target 7 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (233.3310 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

```
Target   8 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0

Target   9 (+) ( 266.6640 millimetres)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0
Target   9 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0

Target  10 (+) ( 299.9970 millimetres)
   1)  0.0          2)  0.0          3)  0.0          4) -1.0
   5) -1.0
Target  10 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0

Target  11 (+) ( 333.3300 millimetres)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0
Target  11 (-)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0

Target  12 (+) ( 366.6630 millimetres)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0
Target  12 (-)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0

Target  13 (+) ( 399.9960 millimetres)
   1) -1.0          2) -1.0          3) -1.0          4)  0.0
   5)  0.0
Target  13 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5)  0.0

Target  14 (+) ( 433.3290 millimetres)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0
Target  14 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0

Target  15 (+) ( 466.6620 millimetres)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5) -1.0
Target  15 (-)
   1) -1.0          2) -1.0          3) -1.0          4) -1.0
   5)  0.0

Target  16 (+) ( 499.9950 millimetres)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0
Target  16 (-)
   1)  0.0          2)  0.0          3)  0.0          4)  0.0
   5)  0.0
```

ACE MANUFACTURING SYSTEM LTD.

AMS
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Machine No :-	M11-719.	Date :-	14:20 Sep 24 2011
Machine Modle :-	MCV-400.	Cond by:-	QAS/DVS.

AXIS CALIBRATION PITCH ERROR REPORT

X				Y				Z			
SL NO				SLNO				SL NO			
100 (X)	0			200 (Y)	0			300 (Z)	0		
101	0	141	0	201	1	241	1	301	1	341	0
102	0	142	0	202	0	242	1	302	0	342	0
103	0	143	0	203	1	243	0	303	0	343	0
104	0	144	-1	204	1	244	0	304	0	344	0
105	0	145	0	205	2	245	3	305	-1	345	0
106	0	146	-1	206	0	246	0	306	0	346	1
107	0	147	0	207	0	247	0	307	0	347	1
108	0	148	1	208	-1	248	0	308	-1	348	0
109	0	149	0	209	1	249	0	309	-1	349	1
110	1	150	0	210	1	250	0	310	-1	350	2
111	0	151	0	211	2	251	0	311	-1	351	0
112	0	152	0	212	1	252	0	312	-1	352	0
113	0	153	-1	213	1	253	0	313	-2	353	0
114	0	154	2	214	0	254	0	314	0	354	0
115	0	155	0	215	0	255	0	315	-1	355	0
116	0	156	0	216	1	256	0	316	0	356	0
117	-1	157	0	217	2	257	0	317	-2	357	0
118	0	158	0	218	0	258	0	318	-1	358	0
119	0	159	0	219	0	259	0	319	-2	359	0
120	0	160	1	220	-1	260	0	320	-1	360	0
121	0	161	0	221	0	261	0	321	-1	361	0
122	-1	162	0	222	1	262	0	322	-1	362	0
123	0	163	0	223	1	263	0	323	-2	363	0
124	0	164	0	224	1	264	0	324	-1	364	0
125	0	165	0	225	0	265	0	325	-1	365	0
126	0	166	0	226	0	266	0	326	-2	366	0
127	0	167	0	227	1	267	0	327	0	367	0
128	0	168	0	228	0	268	0	328	-1	368	0
129	-1	169	0	229	1	269	0	329	-1	369	0
130	0	170	0	230	0	270	0	330	0	370	0
131	0	171	0	231	0	271	0	331	0	371	0
132	0	172	0	232	1	272	0	332	-1	372	0
133	0	173	0	233	0	273	0	333	0	373	0
134	0	174	0	234	1	274	0	334	0	374	0
135	-1	175	0	235	1	275	0	335	0	375	0
136	0	176	0	236	0	276	0	336	0	376	0
137	-1	177	0	237	0	277	0	337	0	377	0
138	0	178	0	238	0	278	0	338	0	378	0
139	-1	179	0	239	1	279	0	339	1	379	0
140	0	180	0	240	0	280	0	340	0	380	0

SL NO	PARARMETER NO	DISCRIPTION	X	Y	Z
1	3620	P COMP REF NO	160	245	350
2	3621	P COMP MIN NO	100	200	300
3	3622	P COMP MAX NO	165	250	355
4	3623	PCOMP MULTIPLICATION	10	10	10
5	3624	P COMP INTERVAL	10	10	10

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

M/c NO :-	M11-719.	COND BY:- QAS/DVS.
Machine Modle :-	MCV-400.	DATE:- 24-Sep-11

AXIS CALIBRATION BALL BAR PARAMETERS

SERVO PARAMETERS RELATED TO CIRCULARITY FOR FANUC SYSTEM

SL No.	PARAMETER No.	TYPICAL VALUE	DESCRIPTION	X	Y	Z	REMARKS
1	1851	###	BACKLASH IN FEED	35	80	40	
2	1852	###	BACKLASH IN RAPID	15	35	25	
3	1825	8000	SERVO LOOP GAIN	5000	5000	5000	
4	1826	500	IN POSITION WIDTH	500	500	500	
5	2021	256	LOAD INERTIA	192	128	161	******
6	2048	600 (###)	BACK ACCEL COMPENSATION VALUE	50	25	50	
7	2094	600 (###)	BACK ACCEL COMPENSATION VALUE	50	25	50	
8	2066	10 TO -20	250 MICRO SEC ACC FEEDBACK	-10	-10	-10	******
9	2068	70 TO700	FEED FWD CO- EFFICIENT	100	100	100	
10	2069	50 TO 200	VELOCITY FEED FWD CO- EFFICIENT FOR RAPID	100	100	100	
11	2071	50 TO 200	TIME DURING WHICH BACK ACCELERATION IS EFFECTIVE	100	100	100	
12	2145	100	VELOCITY FEED FWD CO- EFFICIENT FOR CUTTING	100	100	100	SHOULD NOT BE CHANGED
13	2144	9900	LOOK AHEAD FFW COEFFICIENT FOR CUTTING	9900	9900	9900	
14	2092	9200	LOOK AHEAD FFW COEFFICIENT FOR RAPID	9200	9200	9200	
15	2082	100	STAGE 2 START / END PARAMETER	100	100	100	
16	2039	100	STAGE 2 ACCELERATION AMOUNT	300	300	300	
17	1622	NA	CUT TIME CONSTANT	64	64	64	Same value for 3 axis

NOTE:-

****** DON'T CHANGE THE VALUE

VARIES FOR ALL MACHINES

Ballbar diagnostics table
M11-719-XY-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:21:18

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸-0.1	◂0.3 µm	0.3 µm	(2%)	(12)
Backlash Y	▲-0.7	▾0.8 µm	0.8 µm	(5%)	(7)
Reversal spikes X	▸-2.5	◂-3.0 µm	3.0 µm	(20%)	(2)
Reversal spikes Y	▲-3.4	▾-4.8 µm	4.8 µm	(32%)	(1)
Lateral play X	▸0.9	◂-1.1 µm	0.6 µm	(4%)	(9)
Lateral play Y	▲0.8	▾0.4 µm	0.7 µm	(4%)	(8)
Cyclic error X	↑1.0	↓1.0 µm	0.9 µm	(6%)	(5)
Cyclic error Y	↑1.2	↓1.3 µm	1.2 µm	(8%)	(3)
Servo mismatch	0.03 ms		0.4 µm	(3%)	(11)
Squareness	-6.1 µm/m		0.9 µm	(6%)	(6)
Straightness X	-0.9 µm		0.5 µm	(3%)	(10)
Straightness Y	0.1 µm		0.0 µm	(0%)	(13)
Scaling mismatch	-2.0 µm		1.0 µm	(7%)	(4)
Cyclic pitch X	38.1000 mm				
Cyclic pitch Y	12.0000 mm				
Scaling error X	57.6 ppm				
Scaling error Y	64.3 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset X	-0.1 µm				
Centre offset Y	-0.3 µm				
Positional tolerance	52.1 µm				
Best fit radius	150.0091 mm				
Circularity	9.4 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-719-XY-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:21:18

RENISHAW

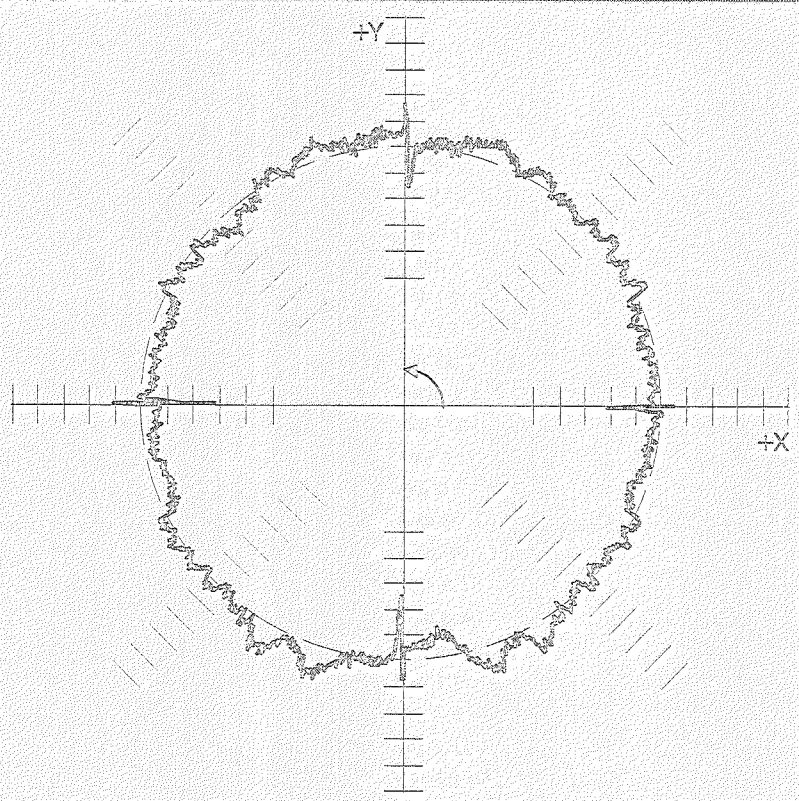
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	8.7µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-719-XY-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:21:18

RENISHAW

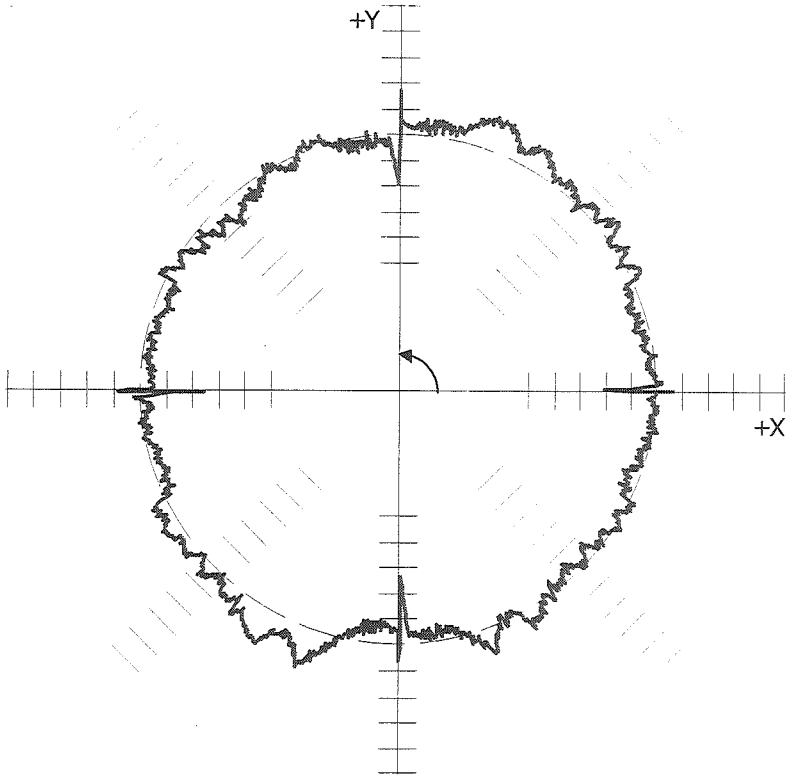
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	8.6µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

2.0 µm/div

Ballbar diagnostics table
M11-719-XZ-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:36:49

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▵0.0	▿0.4 µm	0.4 µm	(2%)	(10)
Backlash X	◃0.0	▹0.3 µm	0.3 µm	(1%)	(11)
Reversal spikes Z	▵-2.5	▿-3.0 µm	3.0 µm	(13%)	(5)
Reversal spikes X	◃-3.5	▹-2.4 µm	3.5 µm	(14%)	(4)
Lateral play Z	▵0.2	▿0.3 µm	0.2 µm	(1%)	(13)
Lateral play X	◃-0.6	▹0.6 µm	0.3 µm	(1%)	(12)
Cyclic error Z	↑0.7	↓0.8 µm	0.6 µm	(3%)	(9)
Cyclic error X	↑1.0	↓0.9 µm	1.0 µm	(4%)	(7)
Servo mismatch	-0.06 ms		0.9 µm	(4%)	(8)
Squareness	26.7 µm/m		4.0 µm	(17%)	(2)
Straightness Z	4.2 µm		2.1 µm	(9%)	(6)
Straightness X	8.0 µm		4.0 µm	(17%)	(1)
Scaling mismatch	-7.6 µm		3.8 µm	(16%)	(3)
Cyclic pitch Z	50.0000 mm				
Cyclic pitch X	45.0000 mm				
Scaling error Z	58.7 ppm				
Scaling error X	84.1 ppm				
Calculated feedrate	1000.8 mm/min				
Centre offset Z	0.6 µm				
Centre offset X	-1.6 µm				
Positional tolerance	66.2 µm				
Best fit radius	150.0107 mm				
Circularity	13.7 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-719-XZ-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:36:49

RENISHAW

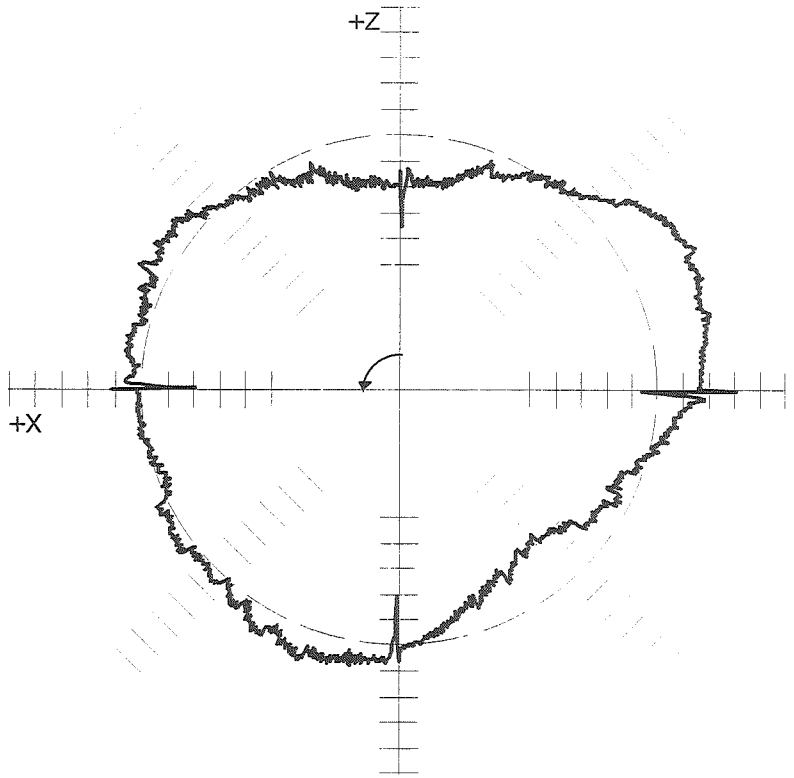
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	13.2µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-719-XZ-B

Operator: Srinivas dv
Date: 2011-Sep-24 17:36:49

RENISHAW

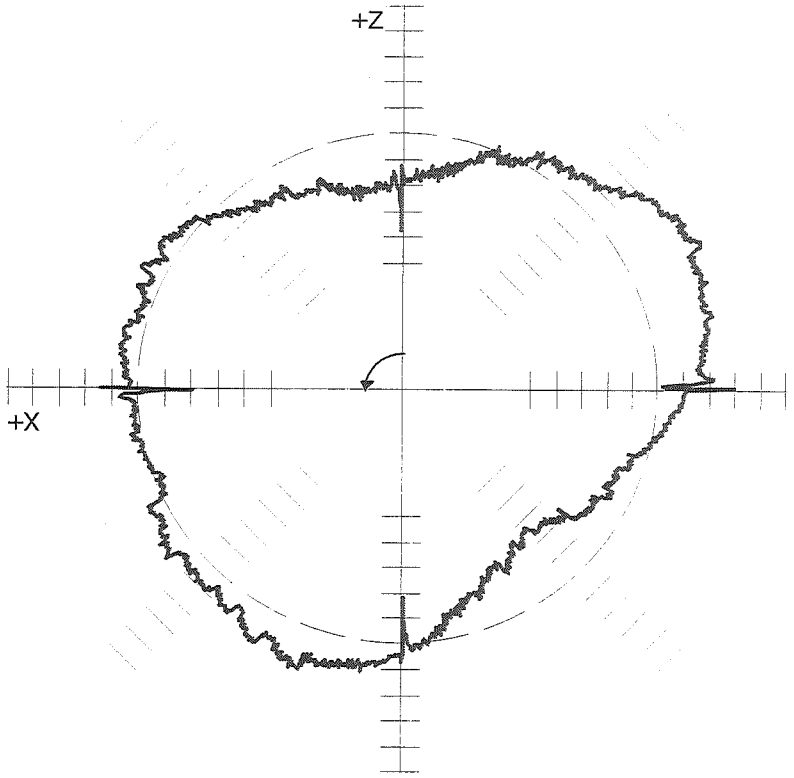
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	13.5µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

2.0 µm/div

Ballbar diagnostics table

M11-719-YZ-A

Operator: Srinivas dv
Date: 2011-Sep-24 17:50:35

RENISHAW

Machine: QuickCheck
Instrument: QC10 ballbar

Error	Magnitude		Independent circularity		Ranking
Backlash Y	›-0.5	‹0.9 µm	0.9 µm	(3%)	(9)
Backlash Z	▵0.0	▿0.3 µm	0.3 µm	(1%)	(13)
Reversal spikes Y	›-3.1	‹-5.0 µm	5.0 µm	(16%)	(2)
Reversal spikes Z	▵-2.3	▿-3.6 µm	3.6 µm	(12%)	(4)
Lateral play Y	›-0.7	‹0.3 µm	0.4 µm	(1%)	(12)
Lateral play Z	▵1.5	▿-0.9 µm	0.8 µm	(3%)	(10)
Cyclic error Y	↑1.3	↓1.5 µm	1.5 µm	(5%)	(7)
Cyclic error Z	↑1.8	↓1.8 µm	1.7 µm	(5%)	(6)
Servo mismatch	0.03 ms		0.4 µm	(1%)	(11)
Squareness	50.8 µm/m		7.6 µm	(25%)	(1)
Straightness Y	5.6 µm		2.8 µm	(9%)	(5)
Straightness Z	2.6 µm		1.3 µm	(4%)	(8)
Scaling mismatch	9.1 µm		4.5 µm	(15%)	(3)
Cyclic pitch Y	12.0000 mm				
Cyclic pitch Z	40.0000 mm				
Scaling error Y	87.3 ppm				
Scaling error Z	57.0 ppm				
Calculated feedrate	1000.4 mm/min				
Centre offset Y	0.6 µm				
Centre offset Z	0.7 µm				
Positional tolerance	71.0 µm				
Best fit radius	150.0108 mm				
Circularity	16.4 µm				

Test Parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:1996(E) Circular Deviation
M11-719-YZ-A

Operator: Srinivas dv
Date: 2011-Sep-24 17:50:35

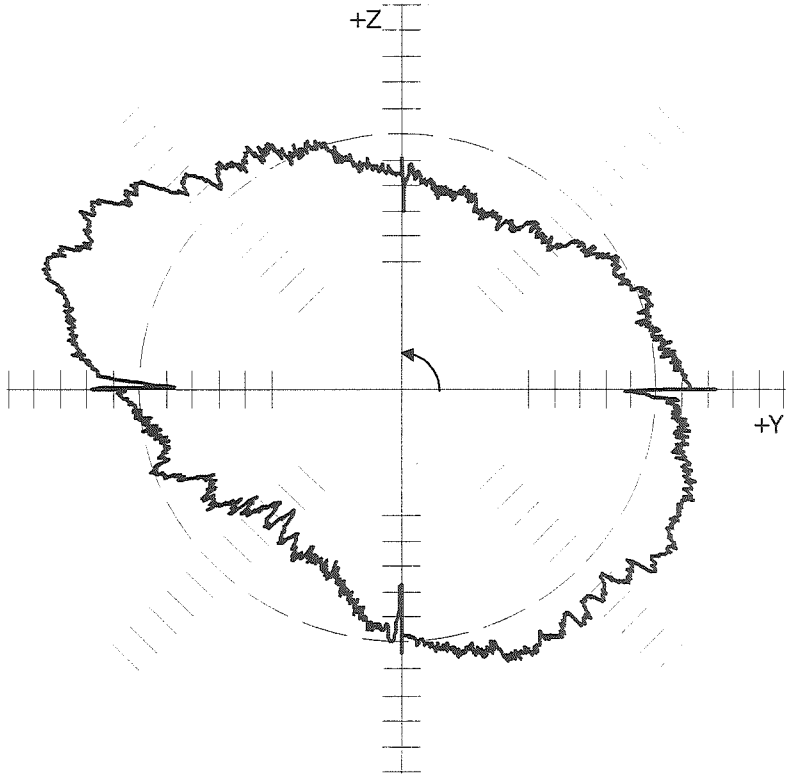
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CW)

Value	16.3µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

2.0 µm/div

ISO 230-4:1996(E) Circular Deviation
M11-719-YZ-A

Operator: Srinivas dv
Date: 2011-Sep-24 17:50:35

RENISHAW

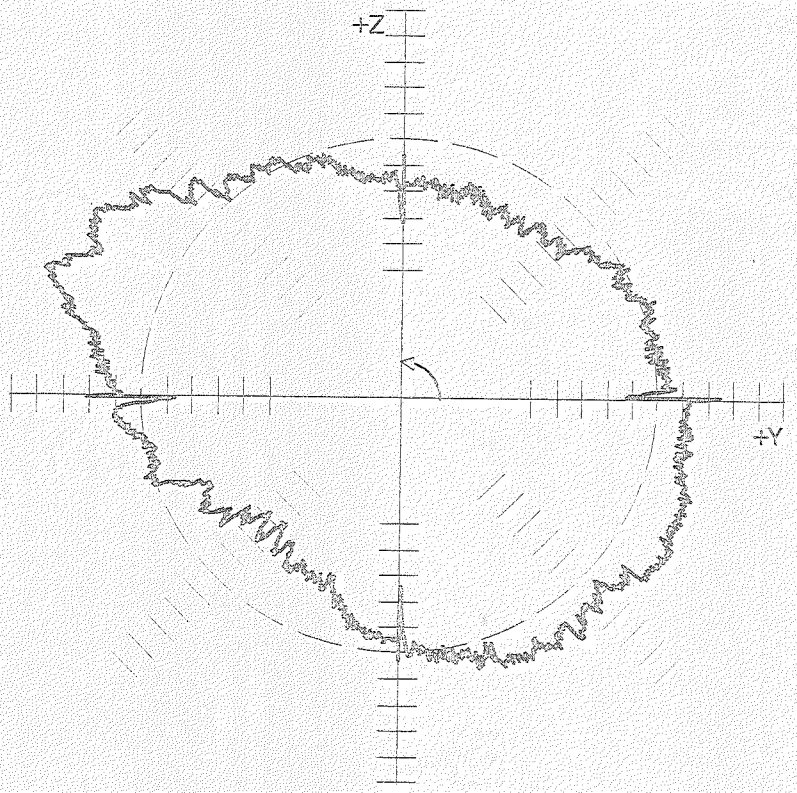
Machine: QuickCheck
Instrument: QC10 ballbar

Circular Deviation (CCW)

Value	15.9µm

Test Parameters

Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

2.0 µm/div